Engineering Physics (2025)
Course code 25PY101-S2
Summative exam version

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M1U1 Plan

CFET

Quantum mechanics

QFET

Opto-electronics

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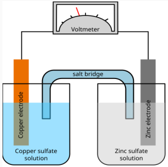
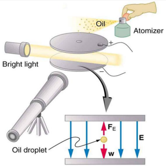
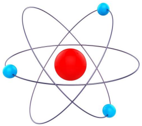


Electrochemical cell



Oil drop experiment

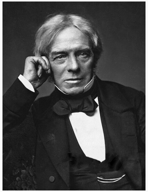
Macroscopic Copper



Microscopic Copper

Estimate: Avogadro number N_A
To electroplate 63.5 g of copper, it takes 2 F of charge. [Hint: 1F (F for Faraday)$=$ 96.485 C, charge of electron $e = 1.602 \times 10^{-19}$ C]

Estimate: Radius of atom
The density of copper is $8.96\,\mathrm{g\,cm^{-3}}$.




Faraday, Millikan

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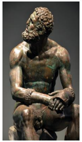
Nature of Metal

(1)

(2)



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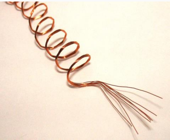


(4), (6)

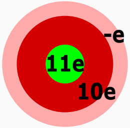
(5)

1. Lustre (Shine)
2. Solid with high 1000 °C melting points
3. Malleable (capable of being shaped)
4. Good electrical conductor
5. Good thermal conductor
6. Ductile (easy to draw wires)

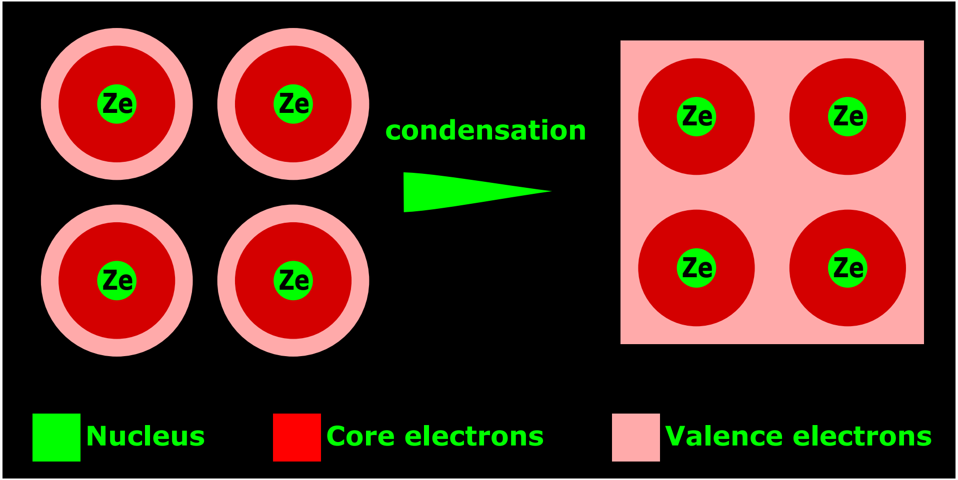
Chemistry

- Metallic bonding
- Screening

Na atom



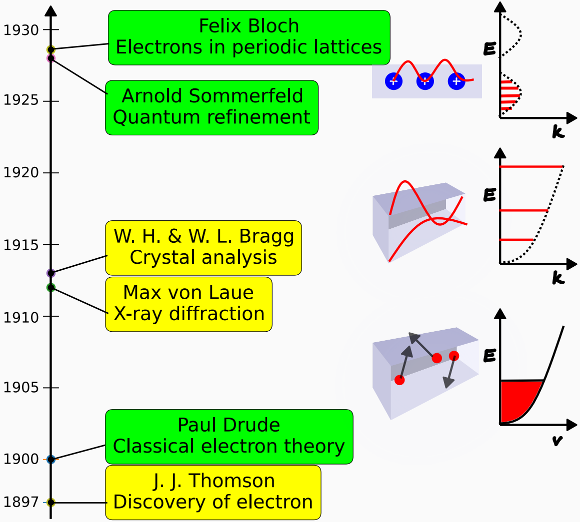
Metallic bonding ↔ electron gas



Valence electrons to electron "gas"

Electron theories of metals

1. Classical free electron theory
2. Quantum free electron theory [M2 U3]
3. Quantum band theory [M2 U3]



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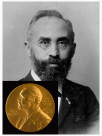
Classical free electron theory

- Ohm's law was discovered in 1827 and needed explanation.
- Proposed by Paul Drude in 1900 using analogy of gas of molecules.
- Mutual repulsion between electrons is neglected i.e. electrons are independent – **independent electron approximation**.
- Assumed "gas" is **free** i.e. not under influence of lattice – **free electron approximation**
- Role of lattice is to redistribute the velocity distribution that obeys **classical** Maxwell-Boltzmann statistics – classical thermodynamics.
- The theory explains Ohm's law
- Hendrik Lorentz extended theory to explain thermal conductivity – hence Drude-Lorentz electron theory.

G. Ohm

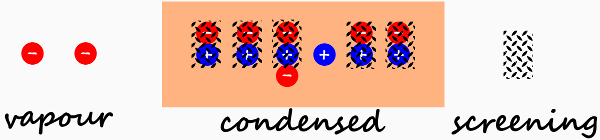
P. Drude



H. Lorentz

Independent electron approximation

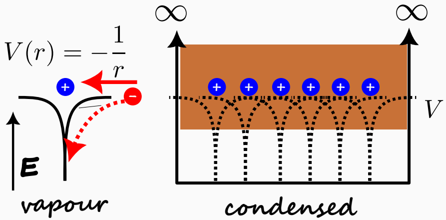
$$V(r) = +\frac{1}{4\pi\epsilon_0}\frac{1}{r} \qquad \cancel{V(r) = +\frac{1}{4\pi\epsilon_0}\frac{1}{r}}$$



- In the condensed phase, there are of the order of 10^{23} electrons.
- The repulsive Coulomb potential between an electron and other electron is **screened** due to large numbers and results in an **independent** electron.
- The problem of motion of $\mathcal{O}(10^{23})$ electrons is reduced to the problem of motion of **single** electron.

Key Insight
The property of single electron determines the properties of electron gas.

Free electron approximation

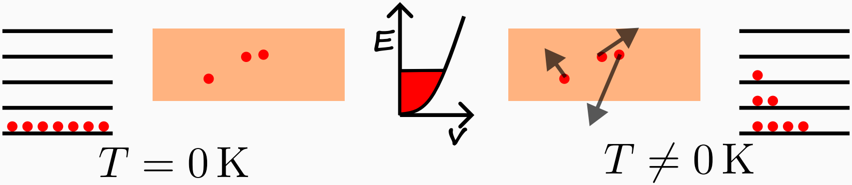


$$V = \begin{cases} 0 & \text{inside} \\ \infty & \text{outside} \end{cases}$$

- The attractive Coulomb potential between ions and an electron is smeared due to large numbers and results in constant potential in the interior of metal with **infinite** barrier at the surface. This potential is called **infinite potential well**.
- This potential traps the electrons analogous to gas atoms in a container.
- The electrons are free to move in the potential well. The kinetic energy K of an electron with mass m and velocity $\mathbf{v}$ is

$$K = \frac{1}{2}mv^2$$

Classical thermodynamics

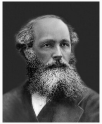


- At temperature $T = 0$, electrons are frozen. At $T \neq 0$, due to the large number of electrons and collisions, the kinetic energy of electrons keeps changing.
- However, the probability to find electron with with lower energy is higher than the probality to find electron with higher energy. This is the essence of Maxwell-Boltzmann statistics.
- The average kinetic energy $\langle T \rangle$ at temperature is

$$\langle K \rangle = \frac{3}{2} k_B T,$$

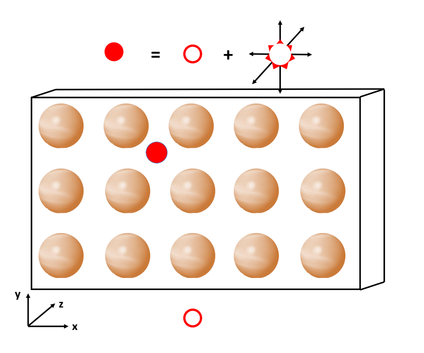
where k_B is the Boltzmann constant.

Boltzmann



Maxwell

Classical free electron theory



- Electron motion is decomposed into **thermal motion** and **drift motion**.
- Thermal motion is due to random collisions with the lattice.
- Drift motion is the net motion of electrons. The electron drifts only between collisions.

Thermal motion

Definition
Mean free path l_{mf} is defined as the average distance the electron travels between collisions.

Definition
Thermal velocity v_{th} is defined as the average velocity of the electron.

- Thermal motion has two properties mean free path l_{mf} and thermal velocity v_{th}.
- Drude assumed that l_{mf} is a material constant.
- At room temperature (300 K), the thermal velocity is of the **order of** $10^5\,\mathrm{m\,s^{-1}}$.

Estimate: Average kinetic energy of electron
Take room temprerature as 27 °C. Express energy in eV. [Hint: $1\,\mathrm{eV} = 1.602 \times 10^{-19}\,\mathrm{J}$]

Drift motion

- In the absence of external force, the drift motion is zero.
- The external force $\mathbf{F}_{ext}$ on the electron is given by the Lorentz force

$$\mathbf{F}_{ext} = -e\mathbf{E} - e\mathbf{v} \times \mathbf{B},$$

  where $\mathbf{E}$ is the electric field and $\mathbf{B}$ is the magnetic field.
- The effect of the collisions is treated as a drag or frictional force

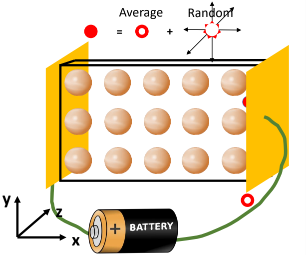
$$\mathbf{F}_{fric} = -m\frac{\mathbf{v}}{\tau}.$$

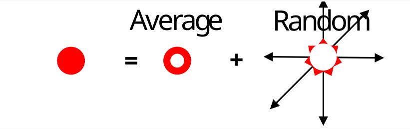
- Since Lorentz extended the theory to incorporate electromagnetic force, the theory is also the Lorentz-Drude theory.

Postulates

1. **Particles:** Electrons are **classical** particles with mass m and charge $-e$.
2. **Kinematics:**
   - **Independent electron approximation:** Electrons are independent and mutual repulsion between them is ignored.
3. **Dynamics:**
   - **Free electron approximation:** Electrons are free and move in an **infinite potential well**.
   - **Thermal motion:** Electron undergoes **random** collision with ion cores. Random collisions lead to thermal motion.
   - **Drift motion:** In between collisions, the Lorentz force acts on electron and leads to drift motion.
4. **Classical Thermodynamics:** The average electron energy is given by Maxwell-Boltzmann statistics.

Ohm’s law





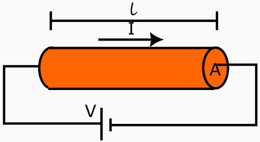
- The motion of electron is decomposed into an **average** part and the **random** part.
- The average part “drifts” with a constant velocity under electric field. The random part has **zero** net motion.

Learning Objectives
Classical electron theory explains Ohm’s law qualitatively.

Ohm's law: Macroscopic version

- The **macroscopic** version of the Ohm's law states that potential V applied across a metal is directly proportional to the current I flowing across it. The constant of proportionaly is called the resistance R.

$$V = IR$$



- Its **microscopic** version relates the electric field and current density
- Electric field $\mathbf{E}$ is defined as the potential per unit distance. Its SI unit is $\mathrm{V\,m^{-1}}$.

$$\mathbf{E} = \frac{V}{\ell}\hat{\mathbf{E}}$$

- Current density $\mathbf{j}$ is defined as the current flowing through a wire per unit area,

$$\mathbf{j} = \frac{I}{A}\hat{\mathbf{j}}$$

  where A is the cross-sectional area of the wire. Its SI unit is $\mathrm{A\,m^{-2}}$.

Ohm's law : Microscopic version

Lets start with the macroscopic version $V = RI$ and derive the microscopic version of Ohm's law

Derivation
Upon substitution of V, I in terms of $\mathbf{E}, \mathbf{j}$, we have

$$\frac{\ell}{V}\mathbf{E} = \frac{A}{I}\mathbf{j} \quad \Rightarrow \quad \mathbf{j} = \frac{\ell}{A}\frac{I}{V}\mathbf{E} \quad \Rightarrow \quad \mathbf{j} = \frac{\ell}{AR}\mathbf{E} = \sigma\mathbf{E}$$

Thus, the microscopic version of Ohm's law is given by

$$\boxed{\mathbf{j} = \sigma\mathbf{E}}$$

where σ is given by

$$\sigma = \frac{\ell}{AR}$$

Conductivity – σ

Definition
The conductivity σ is defined as the current density carried by the wire per unit of electric field.

$$\sigma = \frac{j}{E} \qquad \text{SI unit} \quad [\sigma] = \left[\frac{\ell}{AR}\right] = \Omega^{-1}\,\mathrm{m}^{-1}.$$

Question
A copper wire of length $L = 2\,\mathrm{m}$ and diameter $d{=}1\,\mathrm{mm}$ is connected across a DC source. When a voltage of $V{=}0.5\,\mathrm{V}$ is applied across the ends of the wire, an electric current of $I{=}22\,\mathrm{A}$ is observed. Using these measurements, find the electrical conductivity σ of copper.

Key Insight
Metals have high conductivity.

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Free electron density – n

Definition

- Electron density n is defined as the number of electrons per unit volume. Its SI unit is m^{-3}.
- For metals, each atom contributes as many electrons as its **valency**. For example, Na has valency 1.

Estimate: Electron density of metals

1. Take the radius r_s of atom is nearly $3a_0$, where $a_0 = 0.529\,\text{Å}$ is the Bohr radius.
2. Deduce n is of the order $10^{22}\,\mathrm{cm}^{-3} = 10^{28}\,\mathrm{m}^{-3}$.
3. For Cu, the experimental observed $n = 8.47 \times 10^{28}\,\mathrm{m}^{-3}$.

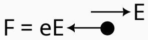
Key Insight

$\frac{r_s}{a_0}$ determines the free electron density.

Drift velocity – v_d

Definition

- **Drift velocity $\mathbf{v_d}$** is defined as the average velocity of an electron in the presence of electric field.
- **relaxation time** τ is defined as the average time between collisions.



Derivation

- *The equation of motion of electron is*
- *During time duration τ, electron reaches drift velocity*

$$\mathbf{a} = -\frac{e}{m}\mathbf{E}$$

$$\boxed{\mathbf{v_d} = -\frac{e\tau}{m}\mathbf{E}}$$

Key Insight

Drift velocity is proportional to the applied electric field.

Mobility – μ

Definition
Mobility μ is defined as the drift velocity per unit electric field

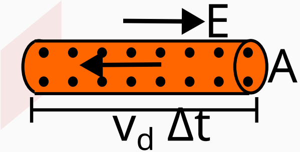
$$\mu = \frac{v_d}{E}.$$

- It is a material property as

$$\mu = \frac{e\tau}{m} \qquad \left[\because \mathbf{v_d} = -\frac{e\tau}{m}\mathbf{E}\right]$$

- Its SI unit is

$$[\mu] = \frac{[v_d]}{[E]} = \frac{[v_d \ell]}{[V]} = \mathrm{m^2\,V^{-1}\,s^{-1}}$$



In a time duration Δt, the number of electrons N in a volume of $V = v_d \Delta t A$ will pass through the cross section of area A.

Derivation

1. *Current I:*
2. *Current density j:*
3. *Conductivity σ:*

$$I = -\frac{Ne}{\Delta t} = -\frac{n \cdot V}{\Delta t} = -nev_d A \tag{1}$$

$$j = \frac{I}{A} = -nev_d = \frac{ne^2\tau}{m} E \tag{2}$$

$$\sigma = \frac{ne^2\tau}{m} = ne\left(\frac{e\tau}{m}\right) = ne\mu \tag{3}$$

Learning Objectives 🎯

We have derived the expression for conductivity.

Relaxation Time – τ

Assumption
Relaxation time τ is assumed to be a **constant**.

Estimate: Relaxation time in metals
Estimate the electron relaxation time in metals.

1. Take conductivity $\sigma \sim 10^7\,\mathrm{S\,m^{-1}}$, number density $n \sim 10^{28}\,\mathrm{m^{-3}}$
2. Take electron mass $m = 9.1 \times 10^{-31}\,\mathrm{kg}$, charge $e = 1.609 \times 10^{-19}\,\mathrm{C}$.
3. Deduce relaxation time $\tau \sim \boxed{10^{-14}\,\mathrm{s} \text{ or } 10\,\mathrm{fs}}$.

- Larger $\tau \rightarrow$ fewer collisions $\rightarrow$ higher conductivity.

Trends

1. As temperature increases, τ decreases. Why?
2. As impurity concentration increases, τ decreases. Why?

Resistivity – ρ

Definition
Resistivity ρ is defined as the inverse of conductivity

$$\rho = \frac{1}{\sigma}$$

Its SI unit is $\Omega\,\mathrm{m}$.

- Resistivity $\rho = \dfrac{1}{\sigma} = \dfrac{m}{ne^2\tau}$.
- Its SI unit is $\Omega\,\mathrm{m}$.

Trends

1. As number density increases, ρ decreases. Why?
2. As temperature increases, ρ increases. Why?

Summary of lecture

- Derived $\sigma = \dfrac{ne^2\tau}{m}$.
- Connected microscopic electron properties to macroscopic Ohm's law.
- Classical free electron theory has qualitative and quantitative agreement with experiment values of conductivity.
- However, the theory has drawbacks. Some of them are
  - Incorrect prediction for conductivity vs valency.
  - Cannot explain anomalous sign of Hall coefficient in some metals.
  - Underestimation of mean free path.
  - Cannot explain classification of materials into conductors. semi-conductors and insulators.
  - Wrong prediction of conductivity vs temperature.
  - Overestimation of heat capacity.

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Conductivity

In a semiconductor, there are two types of charge carriers – electrons and holes. Each carrier is associated with corresponding mobility. Thus, the mobility of electrons is μ_e and that of holes is μ_h.

From the classical free electron theory, the conductivity in terms of mobility and charge density is given by

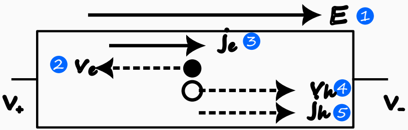
$$\sigma = ne\mu$$

In semiconductors, there are **two channels** for conduction. Thus, the conductivity adds up and is given by

$$\sigma = ne\mu_e + pe\mu_h$$

Key Insight
Both electrons and holes contribute to conductivity.

Drift motion of electrons and holes



- Under electric field, electron and hole drift motion is **anti-parallel**.

$$\mathbf{F}_e = -q_e \mathbf{E}, \qquad \mathbf{F}_h = +q_h \mathbf{E}.$$

  - The sign of their charges are opposite.
  - Electron and hole drift current direction is **parallel**.
  - The total drift current density is given by $\mathbf{j}_{\mathrm{drift}} = \mathbf{j}_e + \mathbf{j}_h$.
  - Drift current is proportional to electric field. $\mathbf{j}_{\mathrm{drift}} = \sigma_e \mathbf{E} + \sigma_h \mathbf{E}$
  - Conductivity is written in terms of carrier density and mobility

$$\sigma_e = ne\mu_e, \qquad \sigma_h = pe\mu_h,$$

  - The drift conductivity is given by

$$\sigma_{\mathrm{drift}} = ne\mu_e + pe\mu_h$$

Intrinsic semiconductor

For the intrinsic case, the number density of electrons equals the number density of holes

$$n = p = n_i$$

Thus, the conductivity is given by

$$\sigma = n_i e\left(\mu_e + \mu_h\right)$$

Carrier density	cm^{-3}
$n_{\mathrm{i-Si}}$	10^{10}
n_{Cu}	10^{22}

Mobility	$\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$
μ_e [Si]	1400
μ_h [Si]	500
μ [Cu]	40

Estimate: Conductivity of intrinsic Si
Compare it with conductivity of copper.

Key Insight
Though the carrier mobility in metals is less than in semiconductors, **very much higher** carrier density leads to higher metallic conductivity.

Extrinsic semiconductor

- n-type: Electrons are **majority** charge carriers

$$n \gg p, \quad \text{so that} \quad ne\mu_e \gg pe\mu_h$$

$$\therefore \sigma \simeq ne\mu_e$$

- p-type: Holes are **majority** charge carriers

$$p \gg n, \quad \text{so that} \quad pe\mu_h \gg ne\mu_e$$

$$\therefore \sigma \simeq pe\mu_h$$

Estimate: Conductivity of extrinsic n-type Si

P doping is $5 \times 10^{16}\,\text{cm}^{-3}$. Compare it with conductivity of intrinsic-Si.

Material	n (in cm^{-3})	σ (in $\text{S}\,\text{m}^{-1}$)
i-Si	10^{10}	10^{-4}
1 ppm n-Si	10^{16}	10^{2}

Key Insight

Doping of the order of 1 ppm increases conductivity of silicon by six orders of magnitude.

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Hall effect

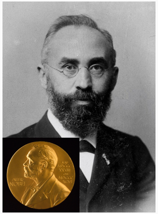
- In the presence of magnetic field acting perpendicular to the flow of electrons, the electrons experience a Lorentz force that deviates their path in the transverse direction. Accumulation of electrons along the edges leads to generation of voltage. This is called **Hall effect**.

E. Hall

- Hall effect is prominent in two dimensional geometry and can be considered the advent of research on thin films.
- It was discovered before the discovery of electron!
- In the context of semiconductors, Hall effect has contribution from holes, in addition to electrons. It also helps determine
  - the type of semiconductor (n or p),
  - the majority carrier concentration, and
  - the majority carrier mobility.

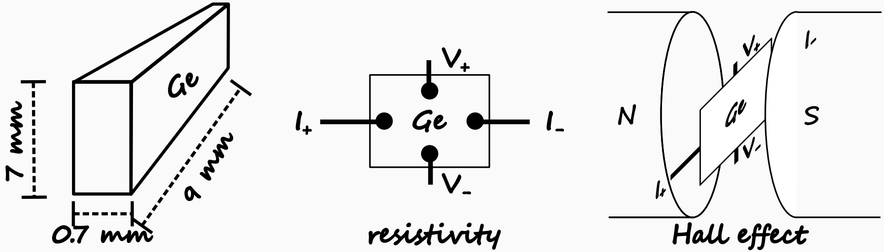
- In the presence of electric field **E** and magnetic field **B**, the electron experiences a force given by

$$\begin{aligned} \mathbf{F}_{\text{Lorentz}} &= \mathbf{F}_{\text{elec}} + \mathbf{F}_{\text{mag}} \\ &= -e\mathbf{E} - e\mathbf{v} \times \mathbf{B} \end{aligned}$$



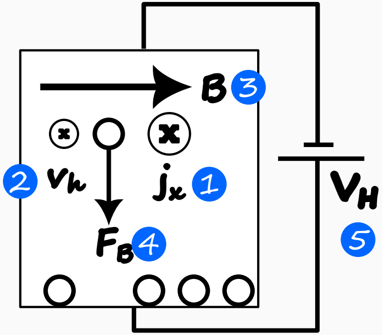
H. Lorentz

Hall effect: experiment



- The aim is to find the charge carrier type of Germanium crystal of dimensions 9 mm x 7 mm x 0.7 mm.
- The resistivity is measured in the **four-probe geometry**.
- Current is applied along x direction.
- Magnetic field of 0.1 T is applied along z direction. [$1\,\mathrm{T}=10^4\,\mathrm{G}$]
- The Hall voltage is measured along y direction.

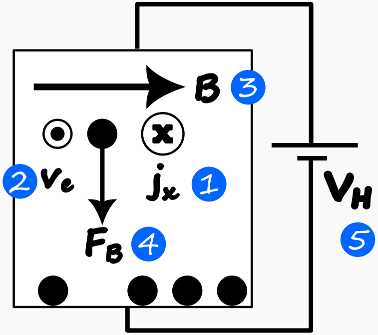
Hall effect in p-type semiconductor



Hall effect for holes

1. Current along x.
2. Electron drift along x.
3. Magnetic field along z.
4. Magnetic force along $-y$.
5. The buildup of holes **induces** an electric field in the $+y$ direction.
6. The corresponding voltage across the width is positive.

Hall effect in n-type semiconductor



Hall effect for electrons

1. Current along x.
2. Electron drift along $-x$.
3. Magnetic field along z.
4. Magnetic force along $-y$.
5. The buildup of electrons **induces** an electric field in the $-y$ direction.
6. The corresponding voltage across the width is negative.

Hall field and Hall voltage

- In the steady state, the magnetic field force is balanced by the induced electric field

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = q(E_y - v_x B_z) = 0$$

- The induced electric field, called the **Hall field** is given by

$$\boxed{E_y = v_x B_z.}$$

- The Hall field produces a voltage across the semiconductor, which is called the **Hall voltage** given by

$$V_H = E_H w$$

where w is the width of the semiconductor.

- The Hall voltage is positive for p-type and negative for n-type semiconductor.

Hall coefficient

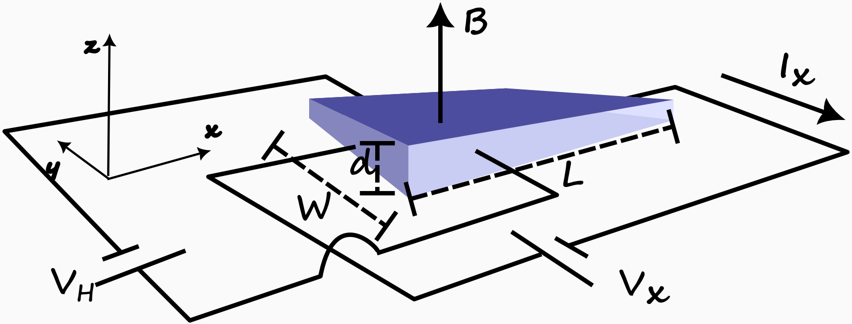
Definition
Hall coefficient R_H is defined as the Hall field per unit current density per unit magnetic field.

- R_H is given by
$$R_H := \frac{E_H}{j_x B_z} = \frac{v}{j_x}$$
- E_H is positive for holes and negative for electrons.
- However, the current density j is related to the drift velocity v by
$$j = nev \qquad [j = pev \quad \text{for holes}]$$
- Therefore,
$$\boxed{R_H(\text{electron}) = -\tfrac{1}{ne}, \qquad R_H(\text{hole}) = +\tfrac{1}{pe}}$$

Charge carrier properties

- Semiconductor type: The sign of Hall voltage determines the type of majority carrier in a semiconductor
- Majority carrier density: The Hall coefficient gives the majority carrier density as
$$\boxed{n = \frac{1}{R_H e}}$$
- Majority carrier mobility: If the conductivity of the semiconductor σ is measured in the absence of magnetic field, then the mobility of majority carriers is given by
$$\boxed{\mu = \frac{\sigma}{ne}}$$

Problem



Determine the majority carrier type, concentration and mobility. $L = 10^{-1}$ cm, $W = 10^{-2}$ cm, $d = 10^{-3}$ cm. $I_x = 1$ mA, $V_x = 12.5$ V, $B = 500$ G, $V_H = -6.25$ mV.

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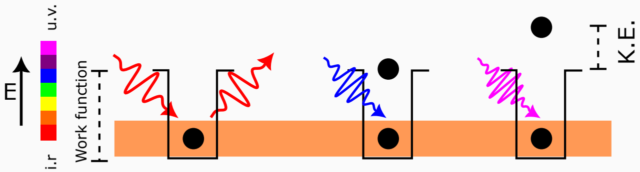
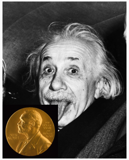
Photo-electric effect

Definition
Light of **suitable** frequency leads to **instantaneous** emission of electron from a metal surface.



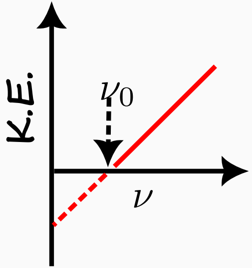


A. Einstein

- Discovered by Albert Einstein in 1905.
- The emitted electron is called the **photo-electron**.
- The minimum energy required to emit an electron is a material property and is called the **work function** of the metal.

Classical explanation of photo-electric effect

- Light is a wave with energy given by its intensity.
- Therefore, even low-frequency light (if sufficiently intense) should eventually provide enough energy to eject electrons — though perhaps after some time delay.



- But it is observed that
  - Emission of electrons is instantaneous – even for low intensity light as long as frequency of light is above a certain threshold ν_0.
  - No electrons are emitted if the frequency is below the threshold, no matter how intense the light is.
  - The kinetic energy of emitted electrons depends on the frequency of light, not its intensity.

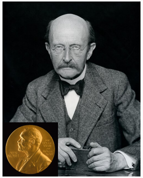
Key Insight
Classical electromagnetic theory cannot explain photoelectric effect!

Quantum mechanical explanation of Photoelectric effect

- Albert Einstein borrowed the idea of **quantum**[1] from Max Planck's theory of blackbody radiation and applied to photoelectric effect.

Planck's law

- Max Planck postulated that the light energy is absorbed or emitted by matter in is a packet of energy. The quantum of energy is related to the frequency ν by $E = h\nu$ where h is the **Planck's constant** given by $6.625 \times 10^{-34}\,\mathrm{J\,s}$.
- The quantum of light energy is called **photon**.



Max Planck

Estimate: Energy of infra-red photon

Wavelength of infra-red photon is $1\,\mu\mathrm{m}$. [Hint: $c = \nu\lambda$]

[1] "Quantum" is Latin for "a certain amount". In the context of physics, the word quantum refers to a **packet**.

Photoelectric effect: energy equation

- Photon with energy $E = h\nu$ transfers energy to the electron, to overcome the work function Φ. The rest of the energy is photoelectron's kinetic energy.

$$\text{K.E.} = h\nu - \Phi$$

- The photoelectrons generate photocurrent.
- In the experiment, the photo electron emitted is decelerated by applying a **stopping potential** V_{sp} till the photocurrent is zero. Thus,

$$\text{K.E.} = eV_{sp}$$

- Therefore, frequency is related to stopping potential as

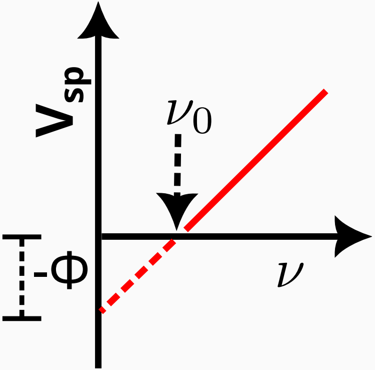
$$eV_{sp} = h\nu - \Phi$$

- The plot of $\nu(\rightarrow x)$ vs $V_{sp}(\rightarrow y)$ is a straight line $y = mx + c$ with slope $m = h/e$ and y-intercept $c = -\Phi$.
- The x-intercept is the minimum frequency of photon to generate photo-current and is called the **threshold frequency** $\nu_0 = \frac{\Phi}{h}$.

Estimate: Energy of X-ray photon
Wavelength of X-ray photon is 0.708 Å.

[Hint: (a). $E \propto 1/\lambda$. (b). $E_{\lambda=1\,\mu m} = 1.24\,\mathrm{eV}$]

Problem



The work function Φ of Au is 4.90 eV.

1. *A blue photon is incident on Au. Does it emit photo-electron? If yes, what is the stopping potential required to have zero photocurrent. [$\lambda_{\text{blue}} = 460$ nm]*
2. *Determine the threshold frequency ν_0 for photon emission of electrons for Au.*

Repeat the above steps for Cs with work function 1.90 eV.

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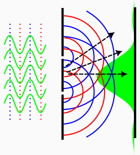
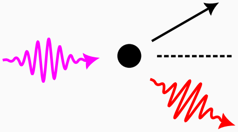
Radiation: Wave vs Particle

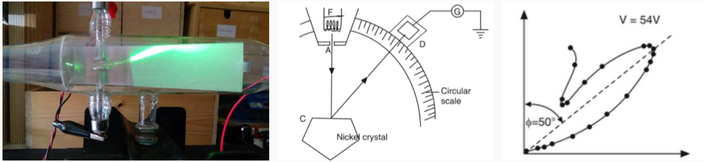


- Young's double slit experiment
- Discovered in 1801
- Light is a wave with amplitude, frequency and wavelength
- **Principle of superposition**: Waves superpose leading to constructive and destructive fringes.
- Animation for double slit experiment [Click]



- Compton effect
- Discovered in 1923.
- Light is a particle with momentum!
- **Conservation of momentum**: Light redshifts (wavelength increases) upon scattering off an electron.

Matter: Particle vs Wave



(a)cathode ray tube, Davison-Germer experiment (b) and data (c).

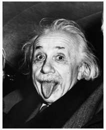
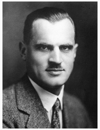
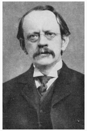
- Thomson's cathode ray tube experiment
- Discovered in 1897
- Electron is a particle

- Davisson and Germer experiment
- Discovered in 1927.
- Electron is a wave!

Key Insight

The name of the game is light-matter interaction. And nature plays the game in a **symmetric** fashion!

Summary of nature of light-matter interaction

	Wave	Particle
Light	 Young's double slit[2]	    Einstein's photo-electric effect Compton's scattering effect
Electron	   Davison's and Germer's diffraction experiment	  Thomson's cathode ray tube experiment

[2] British polymath who has been described as "The Last Man Who Knew Everything" and disproved Newton's corpuscular theory of light.

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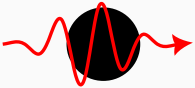
de Broglie hypothesis
Any moving particle is associated with a wave. The waves associated with particles are called **de Broglie waves** or **matter waves**.



- Louis de Broglie in 1924 postulated the existence of **matter waves**.
- He suggested since light wave exhibits particle like behaviour, matter particles should be expected to show wave-like properties. This is the hypothesis of **wave-particle duality**.
- The wavelength λ of the matter wave is related to the momentum p of the matter particle by

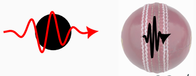
$$\lambda = \frac{h}{p}$$

Matter wave: properties

- Matter waves are due to motion of particle and are independent of charge. Therefore, they are neither electromagnetic waves nor acoustic waves. They are a new kind of waves.
- Can propagate/travel through vacuum and do not require any medium for propagation.
- Smaller the mass, longer the de Broglie wavelength.
- Smaller the velocity, longer the de Broglie wavelength.
- Velocity of matter waves depends on velocity of particle and is not a constant quantity.

Estimate: de Broglie wavelength of thermal electron
Estimate at room temperature.

Estimate: de Broglie λ of macroscopic particle
Cricket ball of mass 150 g is bowled at 140 km h^{-1}.



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Heisenberg uncertainty principle

Uncertainty principle/hypothesis
Precise values of position and momentum of a particle cannot be determined **simultaneously**. If the uncertainty in position is Δx and uncertainty in momentum is Δp, then

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}, \qquad \text{where} \quad \hbar = \frac{h}{2\pi}$$

where $\hbar$ is the reduced/modified Planck's constant.

- Werner Heisenberg postulated the uncertainty principle in 1927.
- In classical mechanics, position and momentum are **deterministic**. In quantum mechanics, they are not deterministic and are associated with **uncertainties**.
- If position is measured with low uncertainty, then measurement of momentum has high uncertainty and vice versa.

Uncertainty → Probability

- The uncertainty principle relates any pair of **conjugate** variables.
- Position – momentum is a pair of conjugate variables.
- Energy – time is also a conjugate variable pair.
- So the uncertainty in energy ΔE is related to the uncertainty in time by Δt by

  $$\Delta E \, \Delta t \geq \frac{\hbar}{2}$$

  This is the second statement/corollary of Heisenberg's uncertainty principle.
- Due to the smallness of reduced Planck's constant, the uncertainty principle is significant for subatomic particles.
- The cause of uncertainty is **limitation of measurement**.

Key Insight 💡
Since the position cannot be measured with certainty, we determine the **probability** of finding an electron at a particular position.

Uncertainty in function of variable

- If uncertainty in variable x is Δx, then the uncertainty in variable $f(x)$ is given by

$$\Delta f = \frac{\mathrm{d}f}{\mathrm{d}x}\Delta x$$

- Kinetic energy E is related to momentum p by

$$E = \frac{p^2}{2m}$$

- If uncertainty in momentum is Δp, then uncertainty in kinetic energy E is given by

$$\Delta E = \frac{\mathrm{d}E}{\mathrm{d}p}\Delta p = \frac{p}{m}\Delta E$$

Problem

- *Uncertainity in position of electron is* $12\,\text{Å}$.
- *Nominal energy of electron is* $16\,\mathrm{e\,V}$.
- *Determine the uncertainty in momentum and kinetic energy*

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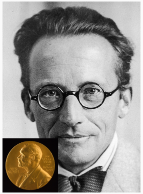
Schrödinger's wave equation

Wave equation hypothesis
If the **wave function** of a particle of mass m is $\Psi(x,t)$, then it satisfies the **wave equation** given by

$$-\frac{\hbar^2}{2m}\cdot\frac{\partial^2\Psi(x,t)}{\partial x^2}+V(x)\Psi(x,t)=j\hbar\frac{\partial\Psi(x,t)}{\partial t}$$

where $V(x)$ is the potential function[a] and j is the imaginary constant $\sqrt{-1}$.



[a] Potential energy and not electric potential difference

- Erwin Schrödinger postulated the wave equation in 1926 that incorporated principles of quanta introduced by Planck and wave-particle duality introduced by de Broglie.
- The wave function $\Psi(x,t)$ describes the behaviour of particle
- $\Psi(x,t)$ can be a **complex** quantity.

$$j\hbar \frac{1}{\boxed{\phi(t)}} \frac{\mathrm{d}\,\boxed{\phi(t)}}{\mathrm{d}t} = \eta$$

- The time dependent side of W.E. is then given by

$$\frac{\mathrm{d}\,\boxed{\phi(t)}}{\mathrm{d}t} = -j\frac{\eta}{\hbar}\boxed{\phi(t)} \qquad \text{related to} \qquad \frac{\mathrm{d}\,y}{\mathrm{d}\,x} = cx$$

- The solution is a sinusoidal wave given by

$$\boxed{\phi(t)} = e^{-j\left(\frac{\eta}{\hbar}\right)t} \qquad \text{related to} \qquad y = e^{-j\omega t}$$

with angular frequency ω given by $[\nu = \frac{\omega}{2\pi}]$

$$\omega = \frac{\eta}{\hbar}, \quad \Rightarrow \eta = \hbar\omega = \frac{h}{2\pi} \cdot 2\pi\nu = h\nu$$

- From Planck's law, $E = h\nu$. Therefore,

$$\eta = E, \qquad \text{and} \qquad \boxed{\phi(t)} = e^{-j\left(\frac{E}{\hbar}\right)t} = e^{-j\omega t}$$

Solution: Time independent part of W.F

$$-\frac{\hbar^2}{2m}\frac{1}{\psi(x)}\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2}+V(x)=\eta$$

- The time independent portion can now be written as

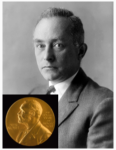
$$-\frac{\hbar^2}{2m}\frac{1}{\psi(x)}\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2}+V(x)=E$$

  where the separation constant is replaced by the total energy E.
- Therefore, the time independent part of W.E. is

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2}+\frac{2m}{\hbar^2}\left(E-V(x)\right)=0$$

- This is called the time independent Schrödinger wave equation. The aim is to solve for the time independent wave function $\psi(x)$ for a given potential $V(x)$.

Physical meaning of wave function



Probability density hypothesis
If the **wave function** of a particle is $\Psi(x, t)$, then the probability density function is given by

$$|\Psi(x, t)|^2$$

- Max Born in 1926 postulated the **interpretation** of wave function.
- Let us assume that we have solved for the time independent part of wave function. Thus,

$$\Psi(x, t) = \psi(x) e^{-j\omega t}$$

- Since $\Psi(x, t)$ is a complex function, it cannot represent physical quantity. Its modulus squared is the **probability density** of finding particle between x and $x + \mathrm{d}x$ and is given by

$$|\Psi(x, t)|^2 = \Psi(x, t) \cdot \Psi^*(x, t) = \psi(x) e^{-j\omega t} \cdot \psi^*(x) e^{+j\omega t} = |\psi(x)|^2$$

- The probability of finding particle between x and $x + \mathrm{d}x$ is then

$$|\psi(x)|^2 \cdot \mathrm{d}x$$

Boundary conditions

- Total probability is unity.

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, \mathrm{d}x = 1$$

  This condition is called **normality condition**.
- Continuity condition
  1. $\psi(x)$ must be finite, single-valued and continuous.
  2. $\frac{\partial \psi(x)}{\partial x}$ must be finite, single-valued and continuous.
- $\psi(x)$ must vanish at infinity.

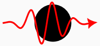
$$\lim_{x \to \infty} \psi(x) = 0, \qquad \lim_{x \to -\infty} \psi(x) = 0$$

  This condition is due to the **physicality condition**.

Problem

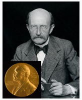
1. *Consider the wave function* $\Psi(x, t) = A \cos\left(\frac{\pi x}{2}\right) e^{-j\omega t}$ *for* $-1 \leq x \leq +3$. *Determine* A *so that* $\int_{-1}^{+3} |\Psi(x, t)|^2 \, \mathrm{d}x = 1$.

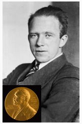
Summary: Postulates of quantum mechanics

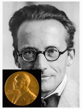


The work of Planck, de Broglie, Heisenberg, Born, and Schrödinger can be summarized as

1. The state of a particle is given by the wave function $\Psi(x, t)$.
2. The probability density of the particle is $|\Psi(x, t)|^2$. The probability of finding the particle in between x and $x + \mathrm{d}x$ is $|\Psi(x, t)|^2 \mathrm{d}x$.
3. If the uncertainty in position is Δx and uncertainty in position is Δp, then $\Delta x \, \Delta p \geq \frac{\hbar}{2}$.
4. The de Broglie wavelength of particle with momentum p is $\lambda = \frac{h}{p}$.
5. The energy E of the particle with frequency ν is $E = h\nu$.







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$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \frac{2m}{\hbar^2}\big(E - V(x)\big)\psi(x) = 0 \qquad \Psi(x,t) = \psi(x)e^{-j\omega t}$$

- The aim is to solve for the time independent wave function $\psi(x)$ for a given potential $V(x)$.
- The potential functions that are useful to describe commonly occurring physical systems are

Applications of Schrödinger's W.E. (contd...)

$V(x)$	Name	Physical systems
0	Constant potential	Particle in free space
$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & \text{otherwise} \end{cases}$	Infinite potential well	Ideal atom
$V(x) = \begin{cases} 0 & -\frac{a}{2} < x < \frac{a}{2} \\ V_0 & \text{otherwise} \end{cases}$	Finite potential well	Work function
$V(x) = \begin{cases} V_0 & -\frac{a}{2} < x < \frac{a}{2} \\ 0 & \text{otherwise} \end{cases}$	Finite barrier	Nuclear fission
		Scanning Tunneling Microscope

Problem
An electron in free space is described by a plane wave given by $\Psi(x,t) = A\exp(j(kx - \omega t))$. *If* $k = 8 \times 10^8\,\mathrm{m^{-1}}$ *and* $\omega = 8 \times 10^{12}\,\mathrm{rad\,s^{-1}}$, *determine*

1. *phase velocity and wavelength of the plane wave*
2. *momentum and kinetic energy (in* eV*)*

Repeat the steps for $k = -1.5 \times 10^9\,\mathrm{m^{-1}}$ *and* $\omega = 1.5 \times 10^{13}\,\mathrm{rad\,s^{-1}}$.

Problem
Determine the wave number, wavelength, angular frequency, and period of wavefunction that describes an electron traveling in free space at a velocity of (a) $5 \times 10^6\,\mathrm{cm\,s^{-1}}$.

Summary of particle in free space

- For a particle in free space, all wave vectors are allowed $k = -\infty$ to $+\infty$.
- The corresponding energy levels are given by

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

- The time independent part of wave function is given by

$$\psi(x) = e^{jkx}$$

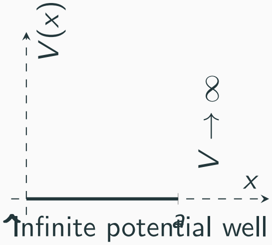
- The time dependent part of wave function is given by

$$\phi(t) = e^{-j\omega t}, \qquad \text{where} \qquad \omega = \frac{E}{\hbar}$$

- The total wave function is given by

$$\Psi_k(x,t) = \psi_k(x)\phi_k(t) = e^{j\overline{kx - \omega t}}.$$

$$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & \text{otherwise} \end{cases}$$



- Consider a particle in a 1-dimensional infinite potential well.
- The well can be considered as a box.

Outside infinite box

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \frac{2m}{\hbar^2}(E - \infty)\psi(x) = 0$$

Inside infinite box

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \frac{2mE}{\hbar^2}\psi(x) = 0$$

$V(x)$: **Infinite potential well – Solution of TISE**

Outside infinite box

- There is no wave function!

$$\psi(x) = 0$$

- Upon applying boundary conditions

Inside infinite box

- The wavefunction is

$$\psi(x) = A\cos kx + B\sin kx$$

Left B.C.

$$\psi(x = 0) = 0$$

$$A\cos kx = 0$$

$$\Rightarrow A = 0$$

Right B.C.

$$\psi(x = a) = 0$$

$$B\sin ka = 0$$

$$\Rightarrow \sin ka = 0$$

- The equation $\sin ka = 0$ is valid if

$$ka = n\pi$$

where n is a positive integer, or $n = 1, 2, 3, \ldots$. The parameter is referred to as quantum number. In terms of the quantum number, the wave vector is given by

$$k = \frac{n\pi}{a}$$

- The prefactor B can be found by applying the **normality condition** of wave function.

$$\int_0^a |B \sin kx|^2 \, \mathrm{d}x = 1, \qquad \Rightarrow \qquad \int_0^a B^2 \sin^2 kx \, \mathrm{d}x = 1$$

$$\therefore B = \sqrt{\frac{2}{a}}$$

- Therefore, the time independent wave function is given by

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

$V(x)$: **Infinite potential well – Energy levels**

- Recall that the wave vector k is related to the energy E by

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \qquad \Rightarrow \qquad E = \frac{\hbar^2 k^2}{2m}$$

- Therefore, the energy is given by

$$E = E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2}$$

Problem

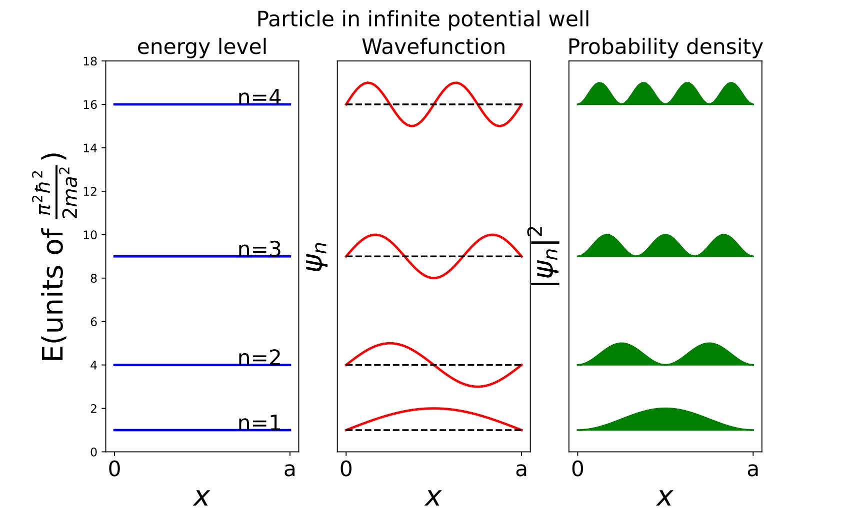
Electron in an infinite potential well of width 5 Å. *Calculate the first three energy levels.*

Key Insight

The energy of particle is **quantized** i.e. it can only have particular **discrete** values.

$V(x)$: **Infinite potential well – Nature of solutions**

- The solutions are **standing** waves.



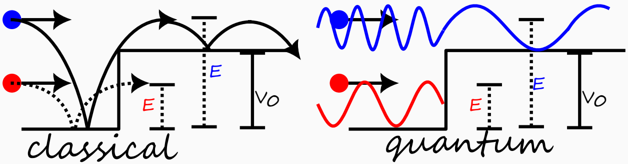
Problem

1. *A particle with mass of* 15 mg *is bound in a 1D infinte potential well that is* 1.2 cm *wide.*
2. *If the energy of the particle is* 15 mJ, *determine the value of n for that state.*
3. *What is the energy of the* $(n+1)$ *state?*
4. *Would quantum effects be observable for this particle?*

Summary of particle in infinite potential well

For a particle in an infinite potential well,

Description	Equation
Discrete wave vectors	$k_n = \frac{n\pi}{a}$.
Energy levels	$E_n = \frac{\hbar^2 k^2}{2m}$, or $E_n = \frac{n^2\hbar^2\pi^2}{2ma^2}$
Time independent part of wavefunction	$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$
Time dependent part of wave function	$\phi_n(t) = e^{-j\omega_n t}$, where $\omega_n = \frac{E_n}{\hbar}$
Total wave function	$\Psi_n(x,t) = \psi_n(x)\phi_n(t)$,



- There are two regions – (I) without barrier and (II) with barrier

Region I

- Total energy E is kinetic energy T_{I} $\qquad T_{\text{I}} = E$
- Wave vector is real $\qquad k_{\text{I}} = \sqrt{\frac{2mT_{\text{I}}}{\hbar^2}}$
- Wave function has sinusoidal form $\qquad \psi_{\text{I}}(x) = A \exp jk_{\text{I}}x + B \exp -jk_{\text{I}}x$

Region II

- There are two cases – (i) total energy is greater than potential energy barrier and (ii) total energy is lesser than potential energy barrier.

Case (i) – $E > V_0$

- Classically, the particle overcomes the potential barrier with decreased kinetic energy.
- Quantum mechanically, the particle overcomes the potential barrier with decreased wave vector.

Region II

- Kinetic energy is reduced by V_0 $\qquad T_{\text{II}} = E - V_0$
- The wave function is given by

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \frac{2m(E - V_0)}{\hbar^2}\psi(x) = 0$$

- Wave vector is real

$$k_{\text{II}} = \sqrt{\frac{2m(E - V_0)}{\hbar^2}}$$

- Wave vector is reduced compared to region I
- Wave function has sinusoidal form $\qquad \psi_{\text{II}}(x) = A\exp jk_{\text{II}}x + B\exp -jk_{\text{II}}x$

Curious case – $E < V_0$

- Classically, the particle **cannot** overcome the potential barrier.
- However, quantum mechanically, it is possible!
- Negative kinetic energy is allowed in quantum mechanics.

Region II

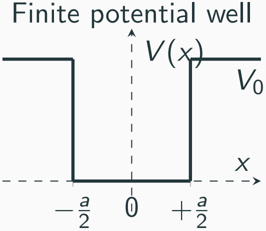
- Total energy E is less than potential barrier
- Kinetic energy is negative! $T_{II} = E - V_0 < 0$
- Wavevector is imaginary

$$k_{II} = \sqrt{\frac{2m(E - V_0)}{\hbar^2}} = j\sqrt{\frac{2m(V_0 - E)}{\hbar^2}} = j\kappa_{II}$$

- Wave function has exponential form

$$\begin{aligned}\psi_{II}(x) &= A\exp jk_{II}x + B\exp -jk_{II}x \\ &= A\exp -\kappa_{II}x + B\exp +\kappa_{II}x\end{aligned}$$

$$V(x) = \begin{cases} 0 & -\frac{a}{2} < x < \frac{a}{2} \\ V_0 & \text{otherwise} \end{cases}$$



- There are three regions – (I) well, (II) right of well, and (III) left of well.
- There are two cases – (i) total energy is less than potential barrier and (ii) total energy is greater than potential barrier
- Let us solve for case (i).

$V(x)$: **Finite potential well – Case (i):** $E < V_0$

Region I

- Kinetic energy is positive so $$\psi_{\mathrm{I}}(x) = A\exp(jk_{\mathrm{I}}x) + B\exp(-jk_{\mathrm{I}}x)$$
- Apply the **symmetry condition** of even or odd wave function

$$\text{Case (1): } A = B \quad \text{or} \quad \text{Case (2): } A = -B$$

Region II

- Kinetic energy is negative so

$$\psi_{\mathrm{II}}(x) = C\exp(-\kappa_{\mathrm{II}}x) + D\exp(+\kappa_{\mathrm{II}}x)$$

- Apply the **finiteness condition**

$$\lim_{x\to\infty} \psi_{\mathrm{II}}(x) = \text{ finite}$$
$$\Leftrightarrow D = 0.$$

Region III

- Kinetic energy is negative so

$$\psi_{\mathrm{III}}(x) = E\exp(-\kappa_{\mathrm{III}}x) + F\exp(\kappa_{\mathrm{III}}x)$$

- Apply the **finiteness condition**

$$\lim_{x\to-\infty} \psi_{\mathrm{III}}(x) = \text{ finite}$$
$$\Leftrightarrow \quad E = 0.$$

- Due to **symmetry** of potential well $\kappa_{\mathrm{III}} = \kappa_{\mathrm{II}}$.

$V(x)$: **Finite potential well – case (1):** $A = B$

- Interface of Region I and Region II i.e. $x = \frac{+a}{2}$

Continuity condition of ψ

$$\psi_{\text{I}}\left(+\frac{a}{2}\right) = \psi_{\text{II}}\left(+\frac{a}{2}\right)$$
$$\Leftrightarrow A\left[\exp\left(j\frac{k_{\text{I}}a}{2}\right) + \exp\left(-j\frac{k_{\text{I}}a}{2}\right)\right]$$
$$= C\exp\left(-\frac{\kappa_{\text{II}}a}{2}\right)$$
$$\Leftrightarrow 2A\cos\frac{k_{\text{I}}a}{2} = C\exp-\frac{\kappa_{\text{II}}a}{2} \qquad (1)$$

Continuity condition of $\frac{\mathrm{d}\psi}{\mathrm{d}x}$

$$\frac{\mathrm{d}\psi_{\text{I}}}{\mathrm{d}x}\left(+\frac{a}{2}\right) = \frac{\mathrm{d}\psi_{\text{II}}}{\mathrm{d}x}\left(+\frac{a}{2}\right)$$
$$\Leftrightarrow jAk_{\text{I}}\left[\exp\left(j\frac{k_{\text{I}}a}{2}\right) - \exp\left(-j\frac{k_{\text{I}}a}{2}\right)\right]$$
$$= -C\kappa_{\text{II}}\exp-\frac{\kappa_{\text{II}}a}{2}$$
$$\Leftrightarrow 2Ak_{\text{I}}\sin\frac{k_{\text{I}}a}{2} = C\kappa_{\text{II}}\exp-\frac{\kappa_{\text{II}}a}{2} \qquad (2)$$

- Dividing (1) by (2) gives the **quantization condition** for wave vector.

$$\boxed{\tan\frac{k_{\text{I}}a}{2} = \frac{\kappa_{\text{II}}}{k_{\text{I}}}}$$

$V(x)$: **Finite potential well – case (1):** $A = B$

$$\tan \frac{k_{\mathrm{I}} a}{2} = \frac{\kappa_{\mathrm{II}}}{k_{\mathrm{I}}} \tag{3}$$

- This is a transcendental type of equation [3] and **cannot** be solved analytically!
- We have to employ numerical techniques.
- First let us introduce dimensionless variable z and constant z_0

$$z = \frac{k_{\mathrm{I}} a}{2} = \frac{a}{2}\sqrt{\frac{2mE}{\hbar^2}}, \qquad z_0 = \frac{a}{2}\sqrt{\frac{2mV_0}{\hbar^2}}$$

  so that (3) is reduced to

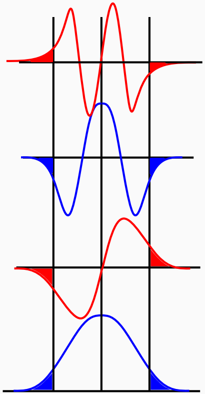
$$z \tan z = \sqrt{z_0^2 - z^2}.$$

- Now let us plot the graph of LHS and RHS of above equation. The points of intersection given you quantized values of z. From this, we can get quantized wave vectors and quantized energies.

[3] Transcendental means beyond conventional knowledge.

$V(x)$: Finite potential well – Nature of solutions

- Wave function penetrates into the finite potential barrier.
- The wave function is either odd or even function.
- The depth of the well determines the number of bound states.
- In the limiting case of very large depth, the penetration tends to zero and the number of bound states tends to infinity leading to particle in infinite potential well.



Key Insight

The wave function penetrates into the finite potential barrier. This is called **Quantum Penetration**.

M2U1 Plan

CFET

Quantum mechanics

QFET

Opto-electronics

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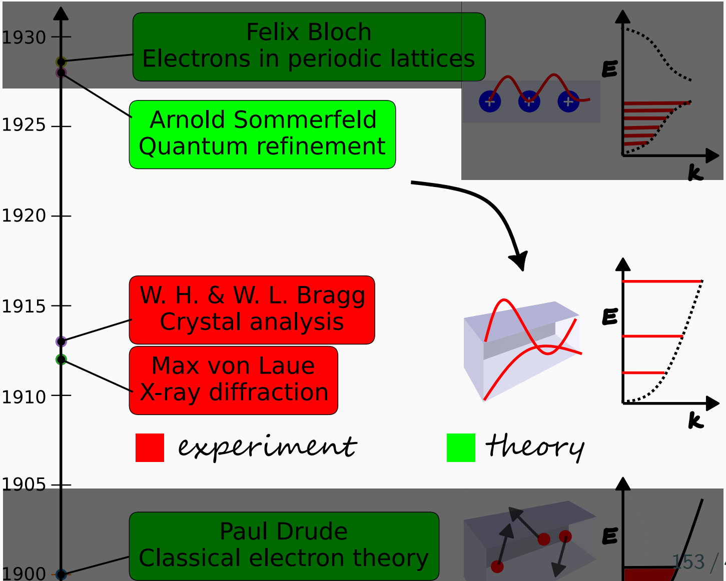
Opto-electronics

Summary of Classical free electron theory (CFET)

- Derived $\sigma = \dfrac{ne^2\tau}{m}$.
- Connected microscopic electron properties to macroscopic Ohm's law.
- Classical free electron theory has qualitative and quantitative agreement with experiment values of conductivity.
- However, the theory has drawbacks. Some of them are
  - Incorrect prediction for conductivity vs valency.
  - Cannot explain anomalous sign of Hall coefficient in some metals.
  - Underestimation of mean free path.
  - Cannot explain classification of materials into conductors. semi-conductors and insulators.
  - Wrong prediction of conductivity vs temperature.
  - Overestimation of heat capacity.

1. Classical free electron theory
2. Quantum free electron theory
3. Quantum band theory



Pauli's exclusion principle

Postulate

- No two electrons can occupy the same quantum state.
- No two electrons can share the same set of quantum numbers.
- Electron obeys Pauli's exclusion principle since it has spin $\frac{1}{2}$.

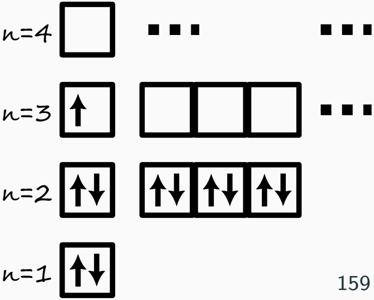
W. Pauli

Definition

A quantum state is defined by a set of quantum numbers.

Problem

What quantum state is occupied by valence electron in Na?



Postulates

1. **Waves:** Electrons are quantum waves with wavevector k, angular frequency ω.
2. **Fermion:** Electron is a spin $\frac{1}{2}$ particle and obeys Pauli's exclusion principle.
3. **Independent electron approximation:** Electrons are independent and mutual repulsion between them is ignored.
4. **Free electron approximation:** Electrons are free and move in an infinite potential well.
5. **Quantum Thermodynamics:** The thermalization is governed by Fermi-Dirac statistics.

Density of states $Z(E) = \dfrac{d(\frac{N}{V})}{dE}$

Definition

- Density of states $Z(E)$ [4] is defined as the rate of change of the number of states per unit volume upto energy E with respect to energy E.
- Therefore, the the number of states per unit volume from energy E to $E + \mathrm{d}E$ is given by

$$Z(E)\,\mathrm{d}E.$$

Number of states per unit volume is

$$\rho_N(E) = \frac{N}{V} = \frac{\pi}{3}\left(\frac{\sqrt{8m}}{h}\right)^3 E^{3/2}$$

[4] In Neamen, density of states is denoted by $g(E)$.

DOS in textbook form

Differentiate $\rho_N(E)$ with respect to E to get the density of states function:

$$Z(E) = \frac{\mathrm{d}\rho_N(E)}{\mathrm{d}E} = \frac{\pi}{2} \cdot \left(\frac{\sqrt{8m}}{h} \right)^3 E^{1/2}$$

$$Z(E) = \frac{\pi}{2} \cdot 8 \cdot \left(\frac{2m}{h^2} \right)^{3/2} E^{1/2}$$

$$\boxed{\therefore Z(E) = 4\pi \cdot \left(\tfrac{2m}{h^2}\right)^{3/2} E^{1/2}.}$$

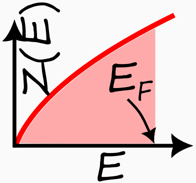
Key Insight
For 3D infinite potential, $Z(E) \propto \sqrt{E}$.

Problem
Find the form of $Z(E)$ for 2D infinite potential.

Definition
Fermi level of a metal is the maximum energy that an electron can have at $T = 0\,\mathrm{K}$.

At absolute zero all states are filled up to the Fermi energy E_F. The total electron density n_c (electrons per unit volume) is

$$
\begin{aligned}
n_c = \frac{N}{V} &= \int_0^{E_F} Z(E)\,\mathrm{d}E = 4\pi \cdot \left(\frac{2m}{h^2}\right)^{3/2} \int_0^{E_F} E^{1/2}\,\mathrm{d}E \\
&= 4\pi \cdot \left(\frac{2m}{h^2}\right)^{3/2} \cdot \frac{2}{3} E_F^{3/2} \\
&= \frac{8\pi}{3} \left(\frac{2m}{h^2}\right)^{3/2} E_F^{3/2}.
\end{aligned}
$$



Relating Fermi energy to electron density

Invert the relation to get the Fermi energy as a function of density:

$$E_F^{3/2} = \left(\frac{h^2}{2m}\right)^{3/2} \cdot \frac{3n_c}{8\pi}.$$

$$\boxed{\therefore\, E_F = \frac{h^2}{2m} \left(\frac{3n_c}{8\pi}\right)^{2/3}.}$$

This is the Fermi energy of a free electron gas in three dimensions at $T = 0$.

Problem
Calculate E_F for metal with electron density $n_c = 5.8 \times 10^{28}\,\mathrm{m}^{-3}$.

Test Your Understanding

1. What are the postulates of QFET.
2. Define Pauli's exclusion principle.
3. Define Fermi energy level.
4. Define density of states function.

Technical terms

- Pauli's exclusion principle
- Fermi level
- Density of states
- Fermi velocity
- Quantum state
- energy level
- degeneracy
- multiplicity

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Quantum thermodynamics: Fermi–Dirac (FD) Statistics

- Describes occupancy of energy states by electrons that obey **Pauli exclusion principle**.
- The probability of occupation of an energy level E at temperature T and Fermi energy level E_F is given by

$$\boxed{P_{FD}(E) = \frac{1}{\exp\left(\frac{E-E_F}{k_B T}\right)+1}.}$$

  Here k_B is the Boltzmann constant equal to $1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$.
- To understand the FD statistics, let us analyze the probability function in two cases –
  1. $T \to 0$ limit
  2. $T \neq 0$

Key Insight
The Fermi-Dirac function is a result of Pauli exclusion principle.

Fermi–Dirac Statistics at the $T \to 0$ limit

- Take limit $T \to 0$ in definition:

$$P_{FD}(E) = \lim_{T \to 0} \frac{1}{\exp\left(\frac{E - E_F}{k_B T}\right) + 1} = \frac{1}{\exp\left(\frac{E - E_F}{0}\right) + 1}.$$

- This limit needs to analyzed for three cases –
  1. $E < E_F$
  2. $E > E_F$
  3. $E = E_F$

Fermi–Dirac Statistics at the $T \to 0$ limit

1. $E < E_F$

$$P_{FD}(E) = \frac{1}{\exp\left(\frac{\text{-ve}}{0}\right) + 1} = \frac{1}{\exp(-\infty) + 1} = \frac{1}{0 + 1} = \frac{1}{1} = 1$$

2. $E > E_F$

$$P_{FD}(E) = \frac{1}{\exp\left(\frac{+\text{ve}}{0}\right) + 1} = \frac{1}{\exp(\infty) + 1} = \frac{1}{\infty + 1} = \frac{1}{\infty} = 0$$

3. $E = E_F$

$$P_{FD}(E) = \frac{1}{\exp\left(\frac{0}{0}\right) + 1}$$

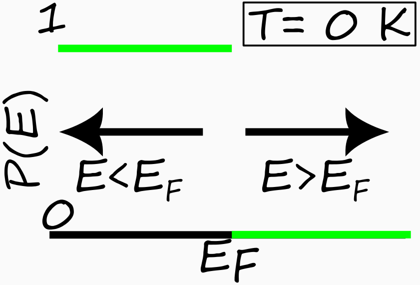
- Since $\frac{0}{0}$ is indeterminate form, $P_{FD}(E)$ is not defined at $T = 0\,\text{K}, E = E_F$.

Fermi–Dirac Statistics at the $T \to 0$ limit

- Hence, at $T = 0\,\mathrm{K}$ the distribution becomes a step function

$$P_{FD}(E) = \begin{cases} 1, & E < E_F \\ 0, & E > E_F \end{cases}$$

where E_F is the Fermi energy.



Key Insight

- At $T \neq 0$, consider $k_B T$ is a small positive quantity $+$ve.

$$P_{FD}(E) = \frac{1}{\exp\left(\frac{E - E_F}{+\text{ve}}\right) + 1}.$$

- This limit needs to analyzed for three cases –[5]
  1. $E \ll E_F$ so that $E - E_F$ is a big negative number i.e. --ve
  2. $E \gg E_F$ so that $E - E_F$ is a big positive number i.e. $++$ve
  3. $E = E_F$

[5] $\gg$ is symbol for very greater than and $\ll$ is symbol for very lesser than

Fermi–Dirac Statistics at $T \neq 0$

1. $E \ll E_F$ [a]

$$
\begin{aligned}
P_{FD}(E) &= \frac{1}{\exp\left(\frac{\text{- -ve}}{+ve}\right) + 1} \\
&\simeq \frac{1}{\exp(-\infty) + 1} \\
&= \frac{1}{0 + 1} \\
&= \frac{1}{1} \\
&= 1
\end{aligned}
$$

$\therefore P_{FD}(E) \simeq 1$

2. $E \gg E_F$

$$
\begin{aligned}
P_{FD}(E) &= \frac{1}{\exp\left(\frac{\text{++ve}}{+ve}\right) + 1} \\
&\simeq \frac{1}{\exp(\infty) + 1} \\
&= \frac{1}{\infty + 1} \\
&= \frac{1}{\infty} \\
&= 0
\end{aligned}
$$

$\therefore P_{FD}(E) \simeq 0$

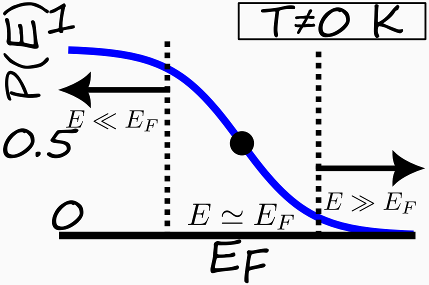
3. $E = E_F$

$$
\begin{aligned}
P_{FD}(E) &= \frac{1}{\exp\left(\frac{0}{+ve}\right) + 1} \\
&= \frac{1}{1 + 1} \\
&= \frac{1}{2}
\end{aligned}
$$

Fermi–Dirac Statistics at $T \neq 0$

- Hence, at $T \neq 0\,\text{K}$ the distribution becomes a "smeared" [6] step function

$$P_{FD}(E) = \begin{cases} \simeq 1 & E \ll E_F \\ \simeq 0 & E \gg E_F \end{cases}$$



Key Insight
At $T = 0\,\text{K}$, the Fermi-Dirac distribution is a "smeared" step function.

[6] smear means smoothened.

Problem
Calculate the probability that an energy level $3k_BT$ above the Fermi energy is occupied by an electron

Problem
Calculate the probability that an energy level $3k_BT$ below the Fermi level is empty

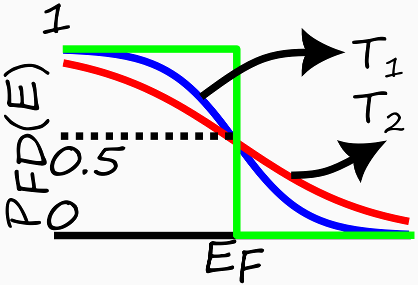
Problem
Fermi energy of a metal is $6.25\,\mathrm{eV}$. Calculate the temperature at which there is a $1\,\%$ probability that a state $0.30\,\mathrm{eV}$ below the Fermi energy level will not contain an electron.

Problem
Calculate the temperature at which the probability that an energy level $0.3\,\mathrm{eV}$ below the Fermi level is occupied is 0.7.

Problem

Calculate the temperature at which the probability that an energy level $3k_B T$ above the Fermi level is occupied is 0.7. *[Hint: This is a wrong question]*

Problem



Statement: $T_1 > T_2$.

Test Your Understanding

1. Explain the variation of Fermi-Dirac distribution function at $T = 0\,\text{K}$ and $T \neq 0\,\text{K}$.
2. Show that the probability of finding an electron of energy ΔE above the Fermi level is same as the probability of not finding an electron at energy ΔE below the Fermi level.
3. Show that the occupancy probabilities of two states whose energies are equally spaced above and below the Fermi energy add up to one.

Technical terms

- occupied level
- unoccupied level
- probability of occupancy
- probability of vacancy

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Specific heat capacity

Definition

- Specific heat capacity [7] C is defined as the rate of change of energy E with temperature T per unit mole of material.

$$C = \frac{\mathrm{d}E}{\mathrm{d}T} \qquad \text{SI unit} \quad [C] = \left[\frac{E}{T} \cdot \frac{1}{\text{mole}}\right] = \mathrm{J\,K^{-1}\,mol^{-1}}.$$

- Metals heat "quickly". When we heat a metal, the heat energy is transferred to electrons. The quickness is measured in terms of heat capacity.

Problem

Specific heat of mercury is $0.14\,\mathrm{J\,g^{-1}\,K^{-1}}$, *water is* $1\,\mathrm{cal\,g^{-1}\,{}^\circ C^{-1}}$, *ethanol is* $2.44\,\mathrm{J\,g^{-1}\,{}^\circ C^{-1}}$. *Which has more specific heat? Which better material for thermometer?* [${}^{200}_{80}Hg$, $1\,\mathrm{cal}=4.2\,\mathrm{J}$, *ethanol* $= CH_3CH_2OH$]

[7] It is also called simply as specific heat.

Specific heat capacity – CFET

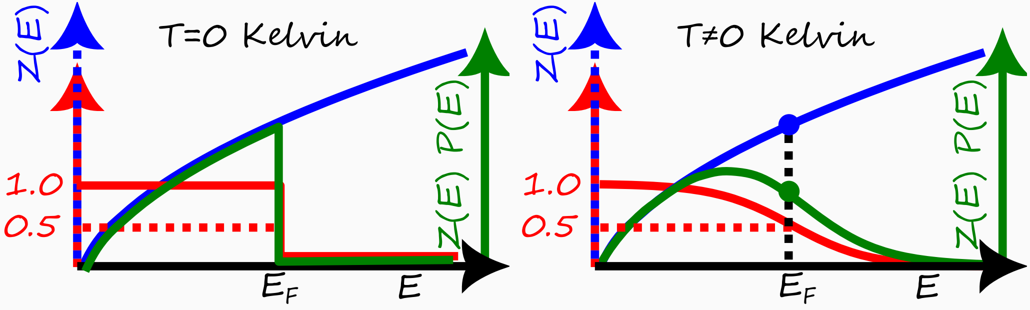
- Energy of electron at temperature T is $E_{\mathrm{el}} = \frac{3}{2} k_B T$.
- Energy of a mole of electrons is

$$E = E_{\mathrm{el}} N_A = \frac{3}{2} k_B N_A T = \frac{3}{2} RT. \qquad \left[R = 8.314\,\mathrm{J\,mol^{-1}\,K^{-1}}\right]$$

- Therefore, specific heat capacity is

$$C_{\mathrm{theory}}^{\text{classical}} = \frac{\mathrm{d}E}{\mathrm{d}T} = \frac{3}{2} R \sim 12\,\mathrm{J\,mol^{-1}\,K^{-1}}.$$

- But experimental values of specific heat capacity $C_{\text{experiment}}$ are **smaller by two orders** of magnitude.



- For the derivation of specific heat capacity, only the **form** of density of states functions is necessary and not the exact expression. For 3D metal, $Z(E) \propto \sqrt{E}$ so that

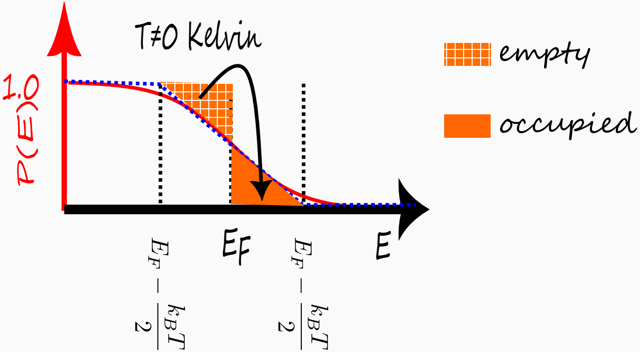
$$Z(E) = \alpha\sqrt{E}, \qquad \text{where } \alpha \text{ is a constant}$$

- At $T = 0\,\mathrm{K}$, electron has energy less than or equal to Fermi energy level.
- The number of electrons per unit volume n_c at $T = 0\,\mathrm{K}$ is given by

$$
\begin{aligned}
n_c &= \int_0^{E_F} Z(E)P(E)\,\mathrm{d}E \\
&= \int_0^{E_F} Z(E)\,\mathrm{d}E \\
&= \int_0^{E_F} \alpha\sqrt{E}\,\mathrm{d}E \\
&= \frac{2}{3}\alpha E_F^{3/2}
\end{aligned}
$$

$$\boxed{\therefore\; n_c = \tfrac{2}{3}\alpha E_F^{3/2}.}$$

Specific heat capacity – Fraction of thermalized electrons



- The number of thermalized electrons is equal to area of the filled triangle multiplied by the density of states in the neighbourhood.
- Area of the filled triangle is

$$A = \frac{1}{2}\frac{k_B T}{2}\frac{1}{2} = \frac{1}{8}k_B T$$

Specific heat capacity – Energy of thermalized electron

- The net increase in energy of thermalized electron is

$$\Delta E_{\text{thermalized}} = E_F + \frac{k_B T}{2} - \left(E_F - \frac{k_B T}{2} \right) = k_B T$$

- Since only a fraction of the electrons are thermalized, the net increase in energy of 1 electron is

$$\begin{aligned} \Delta E_{1\text{ el}} &= \Delta E_{\text{thermalized}} \cdot f_{\text{thermalized}} \\ &= \frac{3}{16} k_B T \cdot \frac{k_B T}{E_F} \end{aligned}$$

- Therefore, the net increase in energy of 1 mole of electrons is

$$\begin{aligned} \Delta E_{1\text{ mole}} &= \Delta E_{1\text{ el}} \cdot N_A \\ &= \frac{3}{16} (k_B N_A) T \cdot \frac{k_B T}{E_F} \\ &= \frac{3}{16} R \frac{k_B T^2}{E_F} \end{aligned}$$

Specific heat capacity – Electronic contribution

- The specific heat capacity is given by

$$
\begin{aligned}
C^{\text{quantum}}_{\text{theory}} &= \frac{\mathrm{d}E_{1\text{ el}}}{\mathrm{d}T} \\
&= \frac{3}{8} R \frac{k_B T}{E_F} \\
&= \frac{3}{2} R \cdot \left(\frac{1}{4} \frac{k_B T}{E_F} \right) \\
&= C^{\text{classical}}_{\text{theory}} \cdot \left(\frac{1}{4} \frac{k_B T}{E_F} \right)
\end{aligned}
$$

$$
\boxed{\therefore\ C^{\text{quantum}}_{\text{theory}} \simeq \left(\frac{k_B T}{E_F} \right) \cdot C^{\text{classical}}_{\text{theory}} .}
$$

- Therefore, quantum theory corrects the classical expression by the fraction $\frac{k_B T}{E_F}$. At room temperature, specific heat capacity is given by

$$
C^{\text{quantum}}_{\text{theory}} = C^{\text{classical}}_{\text{theory}} \cdot \frac{k_B T_{300\,\text{K}}}{E_F} \simeq 0.01\, C^{\text{classical}}_{\text{theory}} \simeq C_{\text{experiment}}
$$

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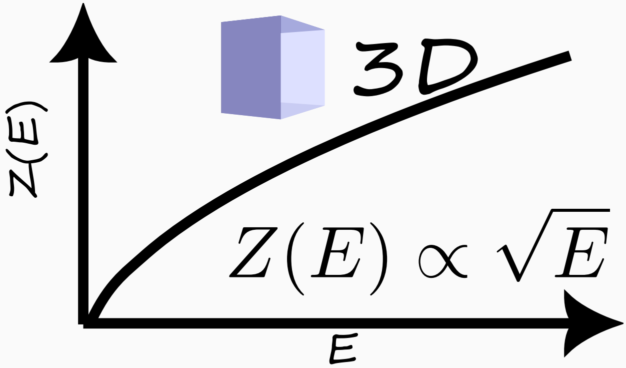
Density of states (3D)

- The form of the DOS in 3D is $Z(E) \propto \sqrt{E}$



Problem

If $Z_{3D}(E) = k\sqrt{E}$, where k is a constant, find the expression for conduction electron number density n_c in terms of Fermi level E_F at $T = 0\,\mathrm{K}$ in a 3D material

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Merits and demerits of QFET

Merits

- Electrical conductivity
- Thermal conductivity
- Heat capacity

Demerits

- Cannot explain classification of condensed matter into metals, semiconductors and insulators
- Occurence of positive Hall coefficient in some metals like Zn.

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Opto-electronics

- In a 1D metal, the electron is free as well as independent.
- For a free electron, the energy is related to the momentum p by

$$E = \frac{p^2}{2m}.$$

- However, since the electron behaves also as a wave, its wavelength λ is related to the momentum by the de Broglie relation

$$\lambda = \frac{h}{p} \quad \Rightarrow \quad p = \frac{h}{\lambda}.$$

- By definition, the wavevector k of the electron is inversely related to the wavelength as

$$k = \frac{2\pi}{\lambda}$$

- Therefore, the energy is related to the wavevector by

$$E = \frac{1}{2m}\left(\frac{h}{\lambda}\right)^2 = \frac{1}{2m}\left(\frac{hk}{2\pi}\right)^2 = \frac{\hbar^2 k^2}{2m}.$$

Summary of Quantum free electron theory (QFET)

- Derived density of states function $Z(E)$.
- Quantum free electron theory addressed the electronic contribution to specific heat of metals
- However, the theory has drawbacks. Some of them are
  - Cannot explain anomalous sign of Hall coefficient in some metals.
  - Cannot explain classification of materials into conductors. semi-conductors and insulators.

Reason for drawbacks

- Invalidity of free electron approximation.
- Drawbacks addressed by Bloch's quantum band theory.

Test Your Understanding

1. If the Fermi level of metal is E_f, what is the fraction of thermalized electrons at temperature T.
2. A thermalized electron according to CFET increases the energy of metal by $\mathcal{O}(k_B T)$. However, according to QFET, the thermalized electron increases the energy of metal by $\mathcal{O}(f k_B T)$, where f is the fraction of thermalized electrons. Explain.
3. Prove that the dispersion relation $E - k$ in metals is given by

$$E(k) = \frac{\hbar^2 k^2}{2m}.$$

4. List the merits and demerits of QFET.

Technical terms

- specific heat capacity C
- ideal gas constant R
- thermalized electron
- $E - k$ dispersion relation

M2U1 Plan

CFET

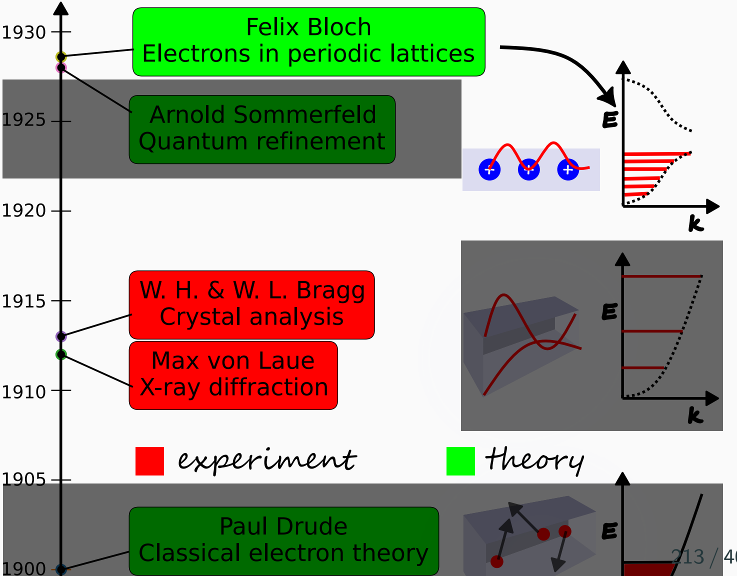
Quantum mechanics

QFET

Opto-electronics

Electron theories of metals

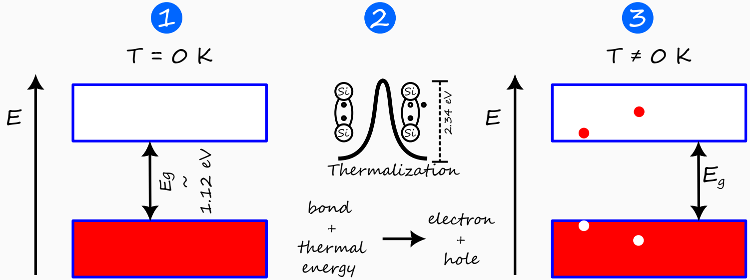
1. Classical free electron theory
2. Quantum free electron theory
3. Quantum band theory (QBT)



Postulates

1. **Waves:** Electrons are quantum waves with wavevector k, angular frequency ω.
2. **Fermion:** Electron is a spin $\frac{1}{2}$ particle and obeys Pauli's exclusion principle.
3. **Independent electron approximation:** Electrons are independent and mutual repulsion between them is ignored.
4. **Periodic potential approximation:** Electrons move in a periodic potential.
5. **Quantum Thermodynamics:** The thermalization is governed by Fermi-Dirac statistics.

Energy band diagram



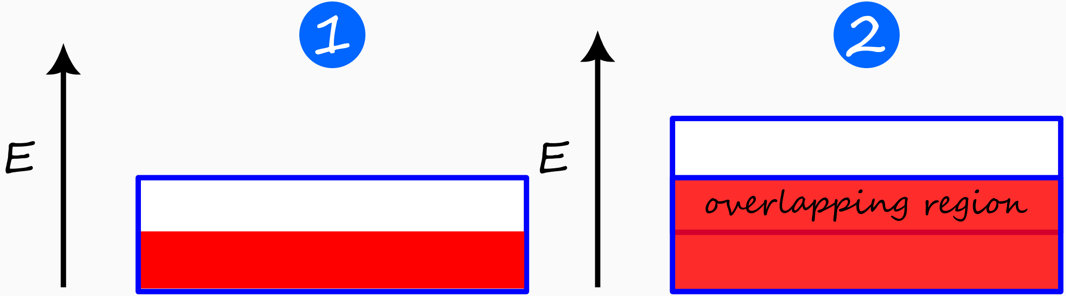
1. At $T = 0\,\mathrm{K}$,
    - The topmost occupied band is called **valence band**.
    - The lower most unoccupied is called **conduction band**.
    - These two bands are separated by a forbidden gap called the **energy band gap**.
2. When temperature is increased, the covalent bond is broken due to thermalization. This results in the formation of **electron-hole pair** [8].
3. At $T \neq 0\,\mathrm{K}$, the electron occupies energy level in the conduction band whereas the hole occupies

Classification of solids

- According to band theory, conductivity of solid is determined by the band gap separating the conduction band and valence band.
- Based on the band gap, the solids are classified into
    1. metals or conductors
    2. semiconductors
    3. insulators

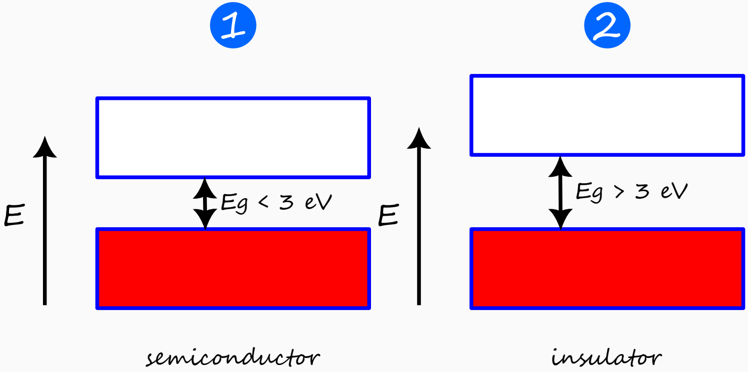
Key Insight
The classification of solids is based on energy band gap.

Energy band diagram – Conductor



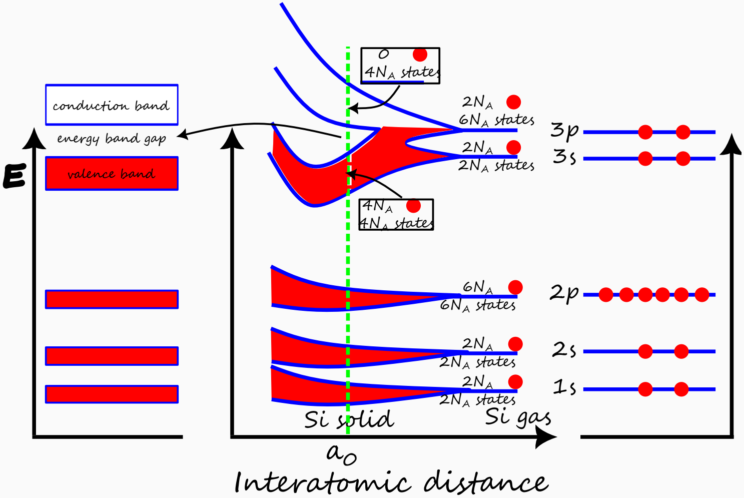
- In conductors, there are two cases
  1. the conduction band is half filled.
  2. the conduction band overlaps with empty upper band.
- In both cases, there is no band gap so that empty states are available for conduction.
- Since there is no band gap, there is no absence of electron at $T \neq 0$. Therefore,

Energy band diagram – Semiconductor



- In either semiconductor or insulator, there is a band gap so that empty states are not available for conduction.
  1. The energy band gap is less than $3\,\mathrm{eV}$ for semiconductor.
  2. The energy band gap is greater than $3\,\mathrm{eV}$ for insulator

Energy band diagram – Si



Key Insight
The valence band is completely filled and the conduction band is completely empty at

Test Your Understanding

1. What are the postulates of QBT?
2. Define the periodic potential approximation.
3. In the parabolic approximation, since the conduction band is approximated by an upward parabola, the conduction band charge carrier is electron-like. Explain.
4. Hole has a negative effective mass. Explain.

Technical terms

- periodic potential approximation
- forbidden energy level
- band structure
- conduction band
- valence band
- energy band gap
- energy band diagram
- conduction band minimum E_c
- valence band maximum E_v
- parabolic approximation
- effective mass of electron and hole

M2U1 Plan

CFET

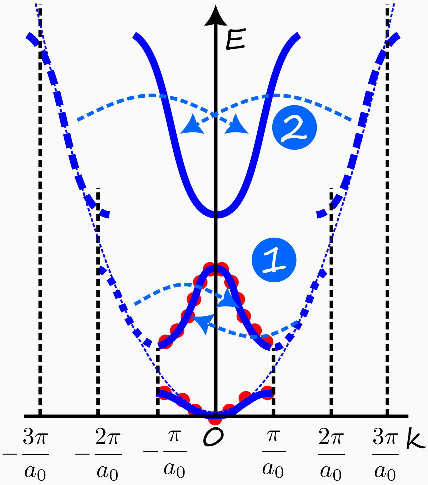
Quantum mechanics

QFET

Opto-electronics

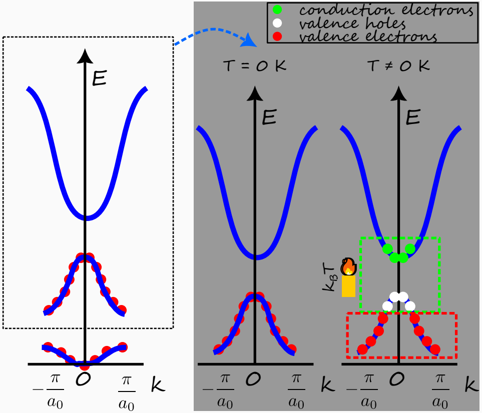
Intrinsic semiconductor

Reduced band dispersion – Band structure



- Due to the periodicity of lattice, the band dispersion can be analyzed within the **zone** from $k = -\frac{\pi}{a_0}$ to $k = +\frac{\pi}{a_0}$.
- The original band dispersion can be reduced to the zone by folding the “arms” of the parabola resulting in the reduced band dispersion. This process is called **zone folding**.
- The reduced band dispersion is called the **band structure** of the material and it is the signature of the material.

Key Insight
Bandstructure is the fingerprint or “DNA” of a crystalline solid.

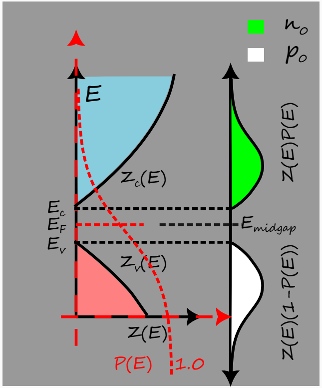


- For a semiconductor, due to the presence of energy band gap, there are no free electrons at $T = 0\,\mathrm{K}$.
- At $T \neq 0\,\mathrm{K}$, the electrons near the valence band maximum thermalize and jump to the levels near the conduction band minimum. This results in quantum vacancy in the valence band. This is called **hole**.
- Conduction electron and valence hole contribute to conduction.

Key Insight

- Hole is the absence of electron in the valence band.
- Hole is not a physical reality.

Electron carrier concentration



- The conduction electron density n_0 is given by

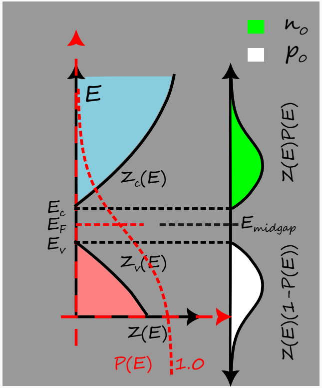
$$n_0 = \int_{E_c}^{\infty} Z_c(E)P(E)\,\mathrm{d}E$$
$$= \int_{E_c}^{\infty} \frac{\alpha^* \sqrt{E - E_c}}{1 + \exp\left(\frac{E-E_F}{k_B T}\right)}\,\mathrm{d}E$$

- By applying Boltzmann approximation and the exact expression for α^*, it can be shown that

$$\boxed{n_0 = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)}$$

where N_c is the **effective density of states** in the conduction band.

Hole carrier concentration



- Similarly the valence hole density p_0 is given by

$$p_0 = \int_{E_c}^{\infty} Z_c(E)\,(1 - P(E))\,\mathrm{d}E$$
$$= \int_0^{E_v} \beta^* \sqrt{E_v - E}\left(1 - \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}\right)\mathrm{d}E$$

- By applying **Boltzmann approximation** and the exact expression for β^*, it can be shown that

$$\boxed{p_0 = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)}$$

where N_v is the **effective density of states** in the valence band.

Intrinsic carrier concentration – $n_0 p_0$ product

- For an intrinsic semiconductor, electron-hole pairs are generated at $T \neq 0\,\mathrm{K}$. So the conduction electron density is equal to valence hole density

$$n_0 = p_0 = n_i$$

  where n_i is defined as the intrinsic carrier concentration.

- The intrinsic carrier concentration is given by the product of electron and hole concentrations

$$\begin{aligned} n_0 p_0 = n_i^2 &= N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) \cdot N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right) \\ &= N_c N_v \exp\left(-\frac{E_c - E_F + E_F - E_v}{k_B T}\right) \\ n_i^2 &= N_c N_v \exp\left(-\frac{E_c - E_v}{k_B T}\right) \end{aligned}$$

Intrinsic carrier concentration

- Therefore, the $n_0 p_0$ is independent of Fermi energy level and is given in terms of the material properties and temperature.

$$\boxed{n_0 p_0 = n_i^2}$$

where n_i is given by

$$\begin{aligned} n_i &= \sqrt{N_c N_v \exp\left(-\frac{E_c - E_v}{k_B T}\right)} \\ &= \sqrt{N_c N_v} \exp\left(-\frac{E_c - E_v}{2k_B T}\right) \\ &= \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right) \end{aligned}$$

where E_g is the energy band gap and is given by

$$E_g = E_c - E_v$$

$$\boxed{\therefore n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{}\right)}$$

Fermi level of intrinsic semiconductor

- Since the electron and hole exist in pairs in an intrinsic semiconductor

$$n_0 = p_0$$
$$\Rightarrow N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

- Taking logarithm on both sides of the equation, we have

$$\ln N_c + \ln\left[\exp\left(-\frac{E_c - E_F}{k_B T}\right)\right] = \ln N_v \ln\left[\exp\left(-\frac{E_F - E_v}{k_B T}\right)\right]$$
$$\Rightarrow \ln N_c - \frac{E_c - E_F}{k_B T} = \ln N_v - \frac{E_F - E_v}{k_B T}$$
$$\Rightarrow \frac{2E_F}{k_B T} = \frac{E_c + E_v}{k_B T} + \ln N_v - \ln N_c$$
$$\Rightarrow E_F = \frac{E_c + E_v}{2} + \frac{k_B T}{2} \ln \frac{N_v}{N_c}$$

- If we substitute the expressions for N_c and N_v, in the ratio of N_c over N_v all the terms cancel out except the effective masses of electron and hole.

Fermi level of intrinsic semiconductor

- The expressions for effective density of states are

$$N_c = 2\left(\frac{2\pi m_n^* k_B T}{h^2}\right)^{3/2}, \qquad N_v = 2\left(\frac{2\pi m_p^* k_B T}{h^2}\right)^{3/2}$$

- Therefore, the Fermi level in terms of effective masses of charge carriers is given by

$$\begin{aligned} E_F &= \frac{E_c + E_v}{2} + \frac{k_B T}{2}\ln\left(\frac{m_p^*}{m_n^*}\right)^{3/2} \\ &= E_{\text{midgap}} + \frac{3k_B T}{4}\ln\frac{m_p^*}{m_n^*} \end{aligned}$$

  where E_{midgap} is defined as the midgap energy level that is in the midpoint of conduction band minimum and valence band maximum and is given by

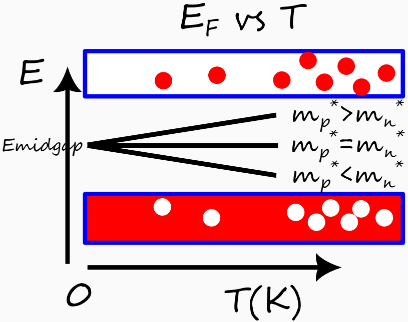
$$E_{\text{midgap}} := \frac{E_c + E_v}{2}$$

$$\boxed{\therefore E_F = E_{\text{midgap}} + \tfrac{3k_B T}{4}\ln\tfrac{m_p^*}{m_n^*}}$$

- Since the Fermi energy level varies with temperature, let us consider two cases – $T = 0\,\text{K}$ and $T \neq 0\,\text{K}$.
- Also, the Fermi energy level varies with effective masses of electron and hole. So let us consider three cases – $m_p = m_n$, $m_p > m_n$, and $m_p < m_n$.

	$T = 0\,\text{K}$	$T \neq 0\,\text{K}$
$m_p = m_n$	E_{midgap}	E_{midgap}
$m_p > m_n$	E_{midgap}	$E_{\text{midgap}} + \delta$
$m_p < m_n$	E_{midgap}	$E_{\text{midgap}} - \delta$

Table 1: Here δ is a positive quantity that linearly increases with temperature.



Key Insight
With temperature, E_F deviates from E_{midgap} for semiconductors with differing electron

Conductivity of intrinsic semiconductor

- Conduction in a semiconductor is through two channels – electron and hole. Therefore

$$\sigma_i = \sigma_e + \sigma_h$$

- For each channel, the conductivity is given by the product of charge carrier concentration and mobility. Therefore

$$\begin{aligned} \sigma_i &= n_0 e \mu_e + p_0 e \mu_h \\ &= n_i e \left(\mu_e + \mu_h\right) \end{aligned}$$

$$\boxed{\therefore \sigma_i = n_i e \left(\mu_e + \mu_h\right)}$$

- If we express the intrinsic carrier concentration in terms of material properties, then

$$\begin{aligned} \sigma_i &= \sqrt{N_c N_v} e \left(\mu_e + \mu_h\right) \exp\left(-\frac{E_g}{2k_B T}\right) \\ &= \sigma_0 \exp\left(-\frac{E_g}{2k_B T}\right) \end{aligned}$$

where σ_0 is a constant in terms of material properties.

Some useful information for calculations

	Si	GaAs	Ge
μ_n ($\mathrm{cm^2\,V^{-1}\,s^{-1}}$)	1350	8500	3900
μ_p ($\mathrm{cm^2\,V^{-1}\,s^{-1}}$)	480	400	1900

Table 2: Mobility of carriers in Si, GaAs, and Ge at 300 K. Taken from Section 5.1 of Neamen

	Si	GaAs	Ge
n_i ($\mathrm{cm^{-3}}$)	1.5×10^{10}	1.8×10^{6}	2.4×10^{13}

Table 3: Intrinsic carrier concentration in Si, GaAs, and Ge at 300 K.

Some useful information for calculations (continued...)

	Si	GaAs	Ge
N_c (cm^{-3})	2.8×10^{19}	4.7×10^{17}	1.04×10^{19}
N_v (cm^{-3})	1.04×10^{19}	7.0×10^{18}	6.0×10^{18}
m_n^*/m_0	1.08	0.067	0.55
m_p^*/m_0	0.56	0.48	0.37

Table 4: Effective density of states and effective mass of charge carriers in Si, GaAs, and Ge . Taken from Section 4.1.3 of Neamen.

	Si	GaAs	Ge
E_g (eV)	1.12	1.42	0.66

Table 5: Energy band gap of Si, GaAs, and Ge at 300 K. Taken from Appendix. B of Neamen.

Problems on Fermi level of intrinsic semiconductor

Problem
A quantum state in the conduction band is $k_B T$ above conduction band minimum E_c of an intrinsic semiconductor. The Fermi level is $0.25\,\mathrm{eV}$ *below E_c. Find the probability that the quantum state is occupied at* $300\,\mathrm{K}$.

Problem
For the same material in above problem, if the effective density of states in conduction band N_c is $2.8 \times 10^{19}\,\mathrm{cm}^{-3}$ *at* $300\,\mathrm{K}$ *, then find the electron concentration n_0.*

Problem
Calculate the position of the intrinsic Fermi level with respect to the center of the bandgap in Si at $300\,\mathrm{K}$.

Problem
Find the resistivity of intrinsic Ge at $300\,\mathrm{K}$. *[Hint: Check useful info slide]*

Problem
The resistivity of intrinsic Si is $2.3 \times 10^3\,\Omega\,\mathrm{m}$ *at* $300\,\mathrm{K}$*. Calculate its resistivity at* $100\,^\circ\mathrm{C}$*.* *[Hint:* $E_g^{Si} = 1.12\,\mathrm{eV}$ *]*

Tough problems

Problem
Calculate the thermal equilibrium electron concentration in Si at $400\,\mathrm{K}$*.* N_c *at* $300\,\mathrm{K}$ *is* $2.8 \times 10^{19}\,\mathrm{cm}^{-3}$*.* *[Hint: Solve the below problem.]*

Problem
Calculate the ratio of conduction electrons at $400\,\mathrm{K}$ *to conduction electrons at* $300\,\mathrm{K}$*.*
[Hint: From the explicit expression for N_c*, observe the form of* N_c *vs* T*.]*

Test Your Understanding

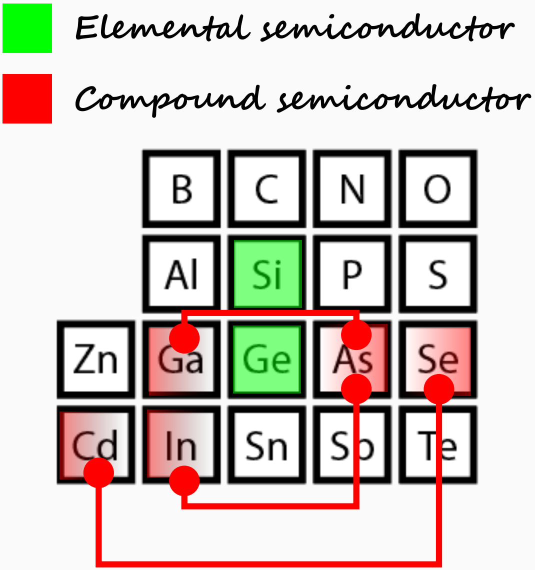
1. Discuss the variation of Fermi level position with temperature.
2. Discuss the variation of conductivity with temperature.

Technical terms

- band structure
- zone folding
- hole
- conduction band effective density of states N_c
- conduction band effective density of states N_v
- intrinsic carrier concentration n_i
- midgap energy level E_{midgap}

Extrinsic semiconductor

Periodic table



Elemental	Compound
Si	GaAs
Ge	InAs
	CdSe

Table 6: Intrinsic semiconductors.

Definition

- Extrinsic semiconductor is a semiconductor with impurities added.
- Depending upon the type of impurities added, extrinsic semiconductor is of two types – n-type and p-type

n-type	p-type
P	Al
As	B
Sb	Ga
In	

Table 7: Dopants for elemental s.c.s.

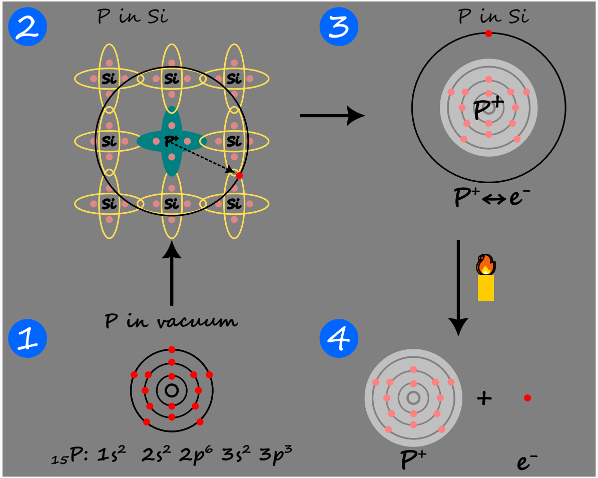
n-type	p-type
Se	Zn
Te	Cd
Si at Ga	Si at As
Ge at Ga	Ge at As

Table 8: Dopants for compound s.c.s.

Problem

Prove that Si doped at Ga position in GaAs acts as a donor dopant.

Donor impurity – Hydrogen-like atom



1. P as isolated atom.
2. P as impurity inside Si.
3. P impurity as $\mathrm{P}^{+} \leftrightarrow \mathrm{e}^{-}$ hydrogen like atom.
4. Due to thermal energy, ionization of P impurity gives free electron.

Key Insight
Doping with donor impurity forms an n-type semiconductor.

Donor impurity – Hydrogen-like atom

Definition

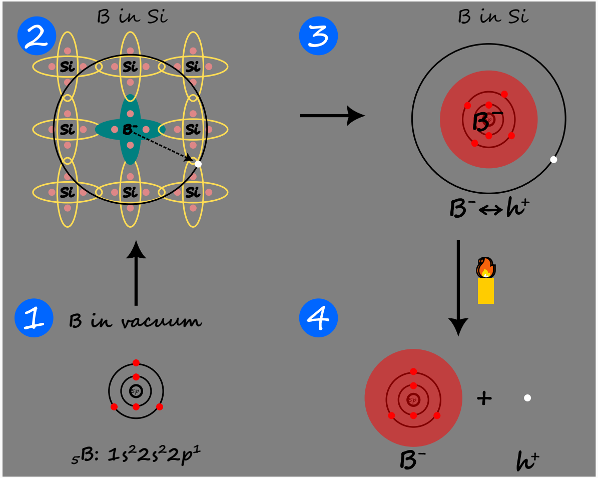
- Donor impurity, also called donor dopant, is an atom that donates an electron to the intrinsic semiconductor upon ionization.

1. Phosphorus in vacuum has 5 electrons in the valence shell.
2. Phosphorus in silicon shares 4 electrons with neighbouring Si atoms for covalent bonding. The **extra** electron is loosely coupled to the phosphorus nucleus.
3. At absolute zero temperature, phosphorus can be modeled as a hydrogenic atom with nucleus as phosphorus ion of charge +1, and a **bound** electron of charge -1 revolving around the nucleus.

$$\mathrm{P}\,\text{in}\,\mathrm{Si} \equiv \mathrm{P}^{+} \leftrightarrow \mathrm{e}^{-}$$

4. When the thermal energy is sufficient to overcome the attractive force between P^{+} and e^{-}, the electron becomes **free** and is released into the Si lattice.

$$\mathrm{P}^{+} \leftrightarrow \mathrm{e}^{-} + \mathcal{O}(k_B T) \rightarrow \mathrm{P}^{+} + \mathrm{e}^{-}$$



1. B as isolated atom.
2. B as impurity inside Si.
3. B impurity as $\mathrm{B}^- \leftrightarrow \mathrm{h}^+$ hydrogen like atom.
4. Due to thermal energy, ionization of B impurity gives free hole.

Key Insight
Doping with acceptor impurity forms a p-type semiconductor.

Acceptor impurity – Hydrogen-like atom

Definition

- Acceptor impurity, also called acceptor dopant, is an atom that accepts an electron from the intrinsic semiconductor upon ionization.

1. Boron in vacuum has 3 electrons in the valence shell.
2. Boron in silicon shares 3 electrons with neighbouring Si atoms for covalent bonding. The 4^{th} electron is absent and can be considered as an **extra** hole that is loosely coupled to the boron nucleus.
3. At absolute zero temperature, Boron can be modeled as a hydrogenic atom with nucleus as Boron ion of charge -1, and a **bound** hole of charge +1 revolving around the nucleus.

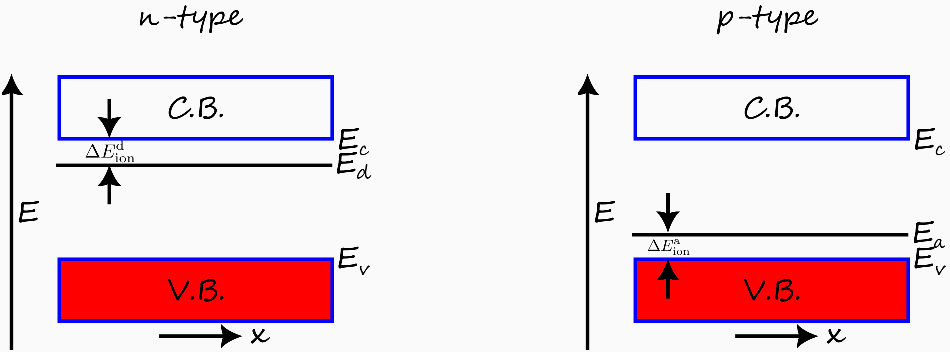
$$\mathrm{B\,in\,Si} \equiv \mathrm{B}^- \leftrightarrow \mathrm{h}^+$$

4. When the thermal energy is sufficient to overcome the attractive force between B^- and h^+, the hole becomes **free** and is released into the Si lattice.

$$\mathrm{B}^- \leftrightarrow \mathrm{h}^+ + \mathcal{O}(k_B T) \rightarrow \mathrm{B}^- + \mathrm{h}^+$$

Definition

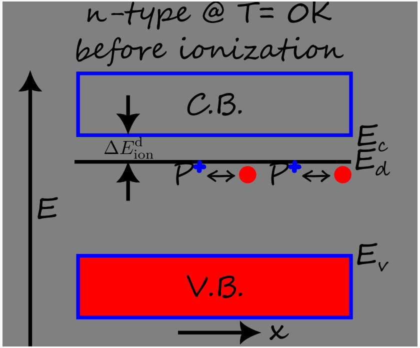
- Donor energy level E_d is the ground state of donor dopant atom.
- Acceptor energy level E_a is the ground state of acceptor dopant atom.

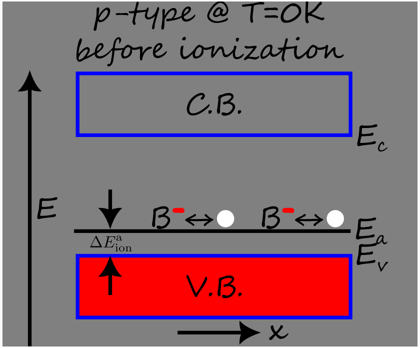


Key Insight

Dopant energy levels – before ionization

- At absolute zero temperature, due to lack of thermal energy, the dopant atoms are **not** ionized.

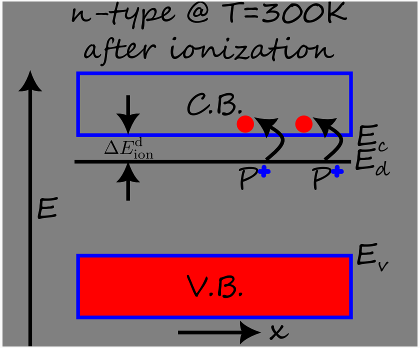


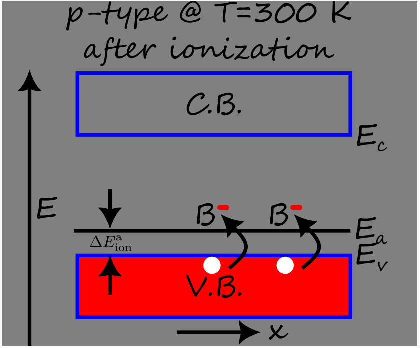


- The electron is **bound** to the donor
- The hole is **bound** to the acceptor

Dopant energy levels – after ionization

- At room temperature, the thermal energy is of the order of ionization energy. This leads to ionization of dopant atoms.

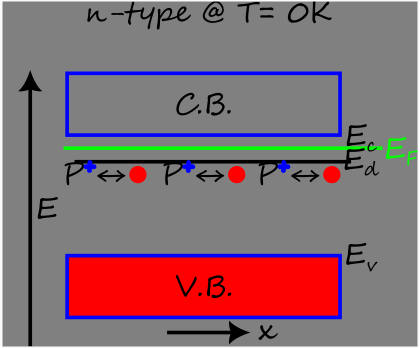


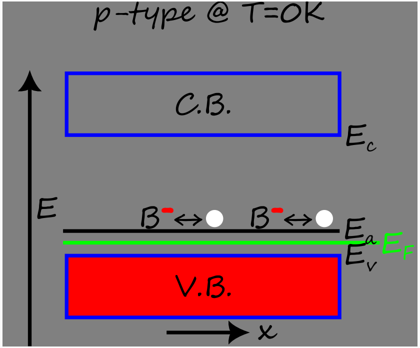


- Upon ionization, the donor dopant releases free electron into the
- Upon ionization, the acceptor dopant accepts electron from valence band

Position of Fermi level – @ 0 K

- At absolute zero temperature, all the energy levels below the Fermi levels must be **occupied** by electrons.





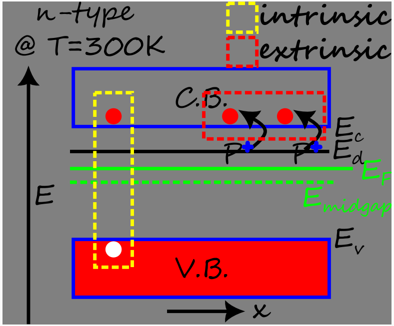
The Fermi energy level lies

- **above** the donor energy level.

The Fermi energy level lies

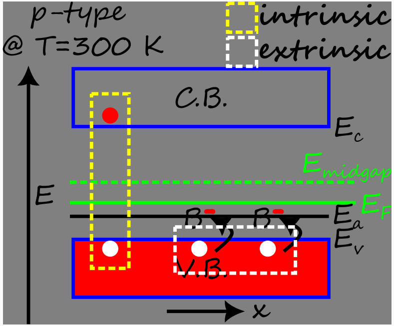
- **below** the acceptor energy level.

- At room temperature, there is contribution to the free carriers from both extr insic and intrinsic



The Fermi energy level lies

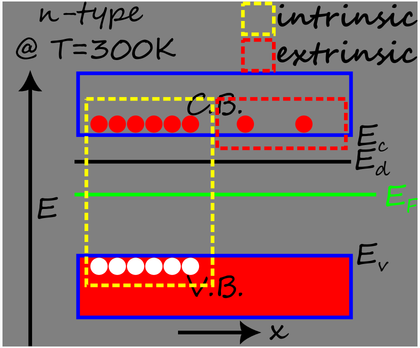
- **below** the donor energy level since donor energy level is **empty**.

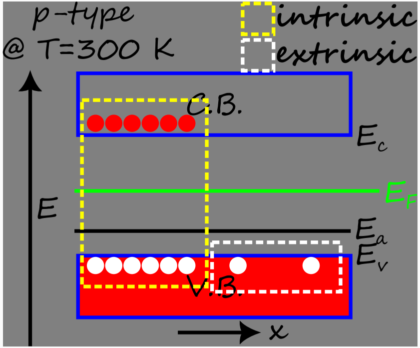


The Fermi energy level lies

- **above** the acceptor energy level since acceptor energy level is **filled**.

- At very high temperature, the contribution to the free carriers from intrinsic **dominates** the contribution from extr insic



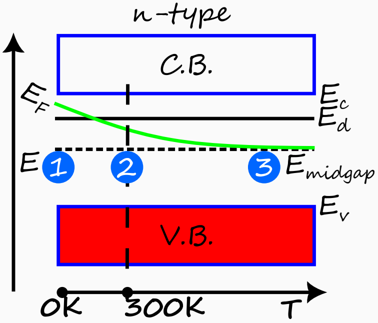


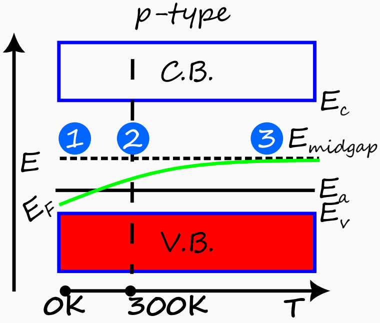
- The Fermi energy level lies near the midgap energy level in both cases.

Key Insight
At very high temperature, extrinsic semiconductor behaves nearly as an intrinsic

Definition

- **Freeze-out** is the condition when the dopants are not ionized.
- **Complete ionization** is the condition when the dopants are completely ionized.





1. At 0 K, the dopants are under freeze-out condition.
2. At 300 K, the thermal energy is nearly equal to ionization energy of dopants

Test Your Understanding

1. What is a donor impurity? What is an acceptor impurity?
2. Differentiate between n-type and p-type semiconductor by sketching their band diagrams.
3. p-type and n-type have more holes and electrons respectively. Explain why they are electrically neutral.
4. What is meant by complete ionization? What is meant by freeze-out?
5. What is the effect of temperature on the working of n-type and p-type semiconductors? Draw energy band diagrams to show the variation in the position of Fermi level with rise in temperature.
6. Draw enery band diagrams for n-type semiconductor at 0 K and at room temperature.
7. Distinguish between extrinsic and intrinsic semiconductors.
8. Sketch a graph of n_0 versus temperature for an n-type semiconductor.
9. Sketch a graph of E_F versus temperature for p-type semiconductor.

Technical terms

- acceptor atoms
- donor atoms
- extrinsic semiconductor
- complete ionization
- freeze-out

End of M2U1

M2U2 Plan

CFET

Quantum mechanics

QFET

Opto-electronics

M2U2 Plan

CFET

Quantum mechanics

QFET

Opto-electronics

pn junction at zero bias

pn junction – at zero bias

Definition

Zero bias condition is considered when external battery is not connected across p-region and n-region.

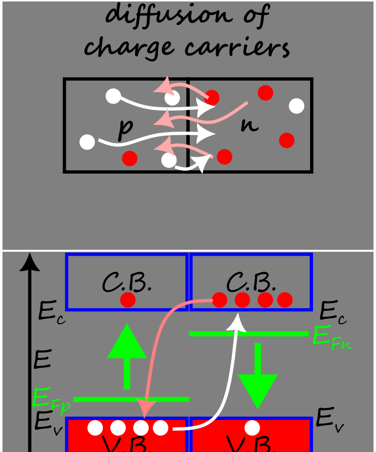
- The word voltage bias is used in the context of electricity to refer to external force.
- At zero voltage bias, there is **no application of external force** on the charge carriers.
- However, there are two internal forces at play –
  1. Diffusion forces due to gradients in concentration of charge carriers.
  2. Drift forces due to internal electric fields.
- The interplay of these internal forces leads to net zero force on charge carriers at thermal equilibrium.

Key Insight

At zero bias, there is interplay of internal drift and diffusion forces.

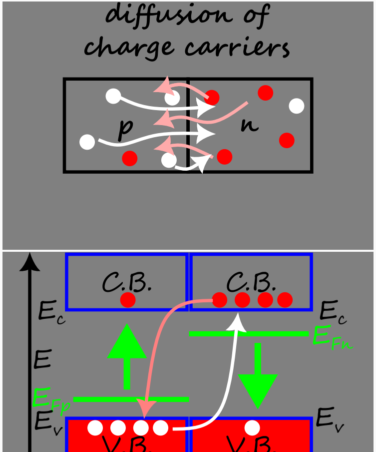
Definition

Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration.



pn junction – Diffusion

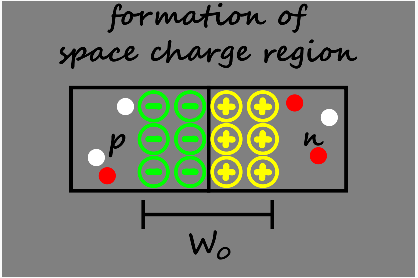
Definition
Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration.



- The difference in Fermi level acts as a “driving force” so that electrons from n-region **diffuse** into the p-region.
- By similar analysis, holes from p-region diffuse into the n-region.
- Diffusion of electrons from n-region lowers its Fermi level E_{Fn}.
- Diffusion of holes from p-region raises its Fermi level E_{Fp}.

Definition

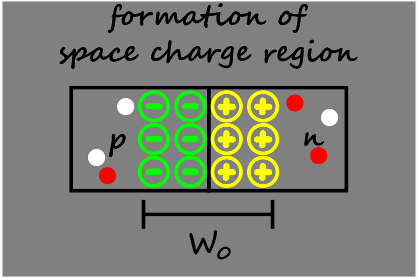
- **Space charge region**, also called the **depletion region**, is the region created as a result of diffusion of free charge carriers across the metallurgical junction.
- The width of the depletion region is called depletion region width W_0.



pn junction at zero bias – Space-charge region

Definition

- **Space charge region**, also called the **depletion region**, is the region created as a result of diffusion of free charge carriers across the metallurgical junction.
- The width of the depletion region is called depletion region width W_0.



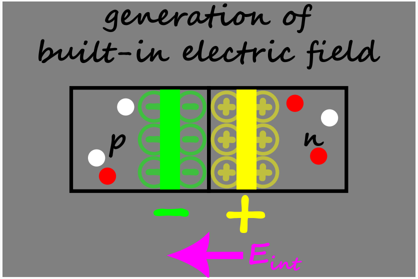
- Due to the diffusion of mobile charge carriers, their **ionic counterparts** are left without partners.
- Space charge region is also called the **depletion region** since the region is depleted [a] of the free charge carriers.

[a] deplete means decrease or remove.

- The depletion region consists of positively charged donor ions on the n-side of the junction and negatively charged acceptor ions on the p-side of the junction.

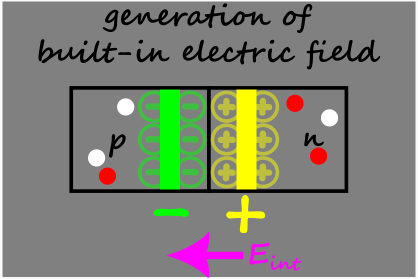
Definition

Built-in electrical field, also called internal electric field E_{int} is the electric field generated due to the oppositely charged regions on either side of the metallurgical junction.



pn junction at zero bias – Built-in electric field

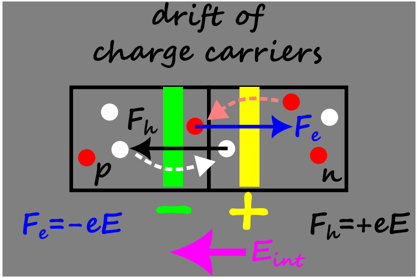
Definition
Built-in electrical field, also called internal electric field E_{int} is the electric field generated due to the oppositely charged regions on either side of the metallurgical junction.



- Within the depletion region, the oppositely charged ions on either side of junction can be considered as plates of a capacitor.
- Since the plates are charged, a **built-in electric field** is generated.
- The built-in electric field is directed from n-region to p-region.

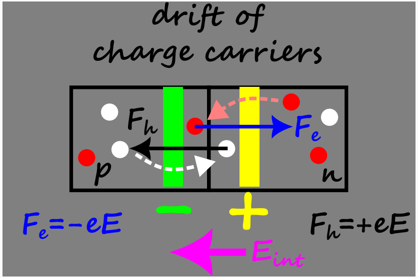
- E_{int} prevents further build up of space charge due to diffusion as it opposes the diffusion of electrons from n-region into p-region and vice versa.

Definition
Drift is the net movement of charged particles due to the application of electric field.



pn junction at zero bias – Drift

Definition
Drift is the net movement of charged particles due to the application of electric field.

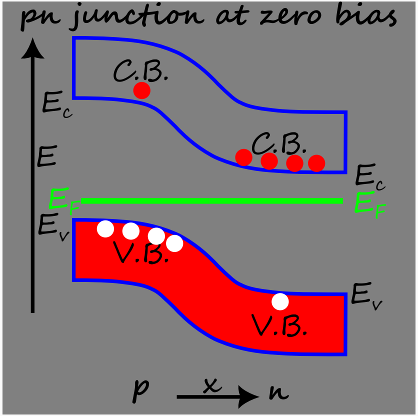


- Consider a diffused electron that is present in the p side of the depletion region.
- Since the built-in electric field is directed from n-region to p-region, the electric force is directed from p-region to n-region.
- Electron drift backs to the n-region.

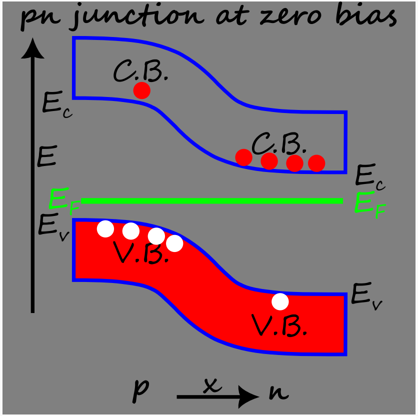
- The diffusion force is counterbalanced by the drift force.
- Similar analysis applies to the diffused hole.

Problem
Prove the counterbalancing effect of drift and diffusion for holes.

pn junction at zero bias – Thermal equilibrium



pn junction at zero bias – Thermal equilibrium



Definition
Thermal equilibrium occurs when the net force on the system is zero.

- Since the difference in Fermi level is a "driving" force, when the system reaches thermal equilibrium, the net force is zero.
- This implies the Fermi level is a constant throughout the system at thermal equilibrium.

- Since the external force is zero at zero bias, the pn junction is at thermal equilibrium.

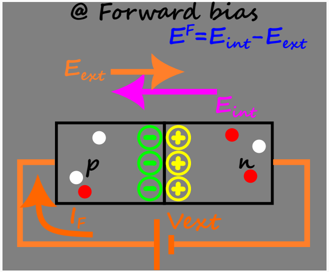
Key Insight
At thermal equilibrium, E_F is a constant.

pn junction at non-zero bias

pn junction – at non-zero bias

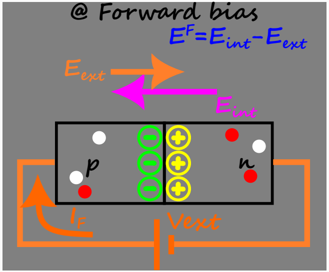
- At non-zero voltage bias, there is **an application of external force** on the charge carriers. As a result, the pn junction cannot attain thermal equilibrium.
- Since there is no thermal equilibrium, the Fermi energy is not a constant across the pn junction.
- Since the direction of built-in electric field is from n-region to p-region, the behaviour of p-n junction is different when the external field E_{ext} is parallel or anti-parallel to the built-in electric field E_{int}.
    1. $E_{\text{ext}} \uparrow\downarrow E_{\text{int}}$ corresponds to forward biasing.
    2. $E_{\text{ext}} \uparrow\uparrow E_{\text{int}}$ corresponds to reverse biasing.

Key Insight
At non zero bias, the Fermi energy is not a constant across the pn junction.



Forward
$V_F > 0$
$E^F < E_{\text{int}}$
$W_F < W_0$
$V_{\text{F}} < V_{\text{int}}$

pn junction – at forward bias



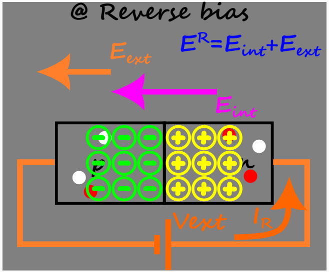
Forward
$V_F > 0$
$E^F < E_{\text{int}}$
$W_F < W_0$
$V_{\text{F}} < V_{\text{int}}$

Definition

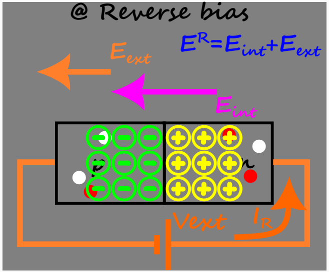
Forward bias condition is considered when the positive electrode of battery with potential V_F is connected to p-region and negative electrode is connected to n-region.

- The external bias $V_{\text{ext}} = V_F$ is from p-region to n-region.
- The external field E_{ext} is from p-region to n-region. The internal field E_{int} is **anti-parallel** to E_{ext}. The net electric field E^F is lower than the case of zero bias.
- So less space charge is created in depletion region. So we have smaller space charge width W_F.

pn junction – at reverse bias



Reverse
$V_R < 0$
$E^{\mathrm{R}} > E_{\mathrm{int}}$
$V_{\mathrm{R}} > V_{\mathrm{int}}$
$E_{\mathrm{Fn}} - E_{\mathrm{Fp}} < 0$



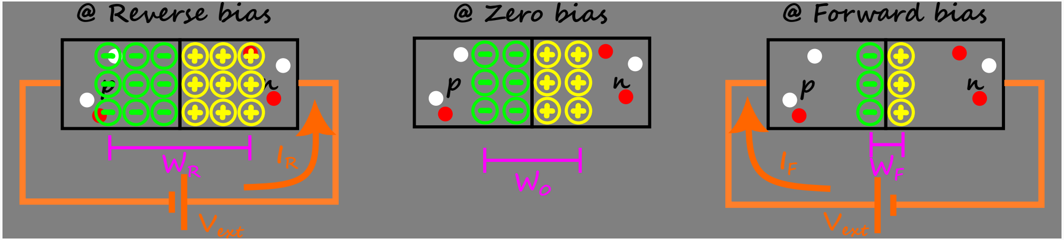
Reverse
$V_R < 0$
$E^{\mathrm{R}} > E_{\mathrm{int}}$
$V_{\mathrm{R}} > V_{\mathrm{int}}$
$E_{\mathrm{Fn}} - E_{\mathrm{Fp}} < 0$

Definition

Reverse bias condition is considered when the positive electrode of battery is connected to n-region and negative electrode is connected to p-region.

- The external bias $V_{\mathrm{ext}} = V_R$ is from n-region to p-region.
- The external field is from n-region to p-region. Under reverse bias condition, the internal field is **parallel** to the external field. The net electric field E^R is higher than the case of zero bias.
- So more space charge is created in depletion region. So we have larger space charge width W_R.

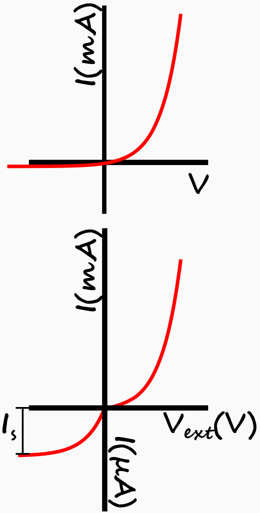
Comparison of depletion region widths at different biases



$$W_R > W_0 > W_F$$

Large forward current vs tiny reverse current

- Under biasing conditions, the electrodes of the battery change the Fermi levels of the p and n regions.
- Since the negative electrode supplies electrons, it raises the Fermi level of the connected region. Similarly, positive electrode lowers the Fermi level.
- Under forward biasing, the energy of electrons at n-region is raised. The flow of electrons to lower energy p-region is possible. Similarly, the energy of holes at p-region is lowered. The flow of holes to higher energy n-region is possible. This leads to **large forward current**.
- Under reverse biasing, the energy of electrons at n-region is lowered. The flow of electrons to higher energy p-region is difficult. Similarly, the energy of holes at p-region is raised. The flow of holes to lower energy p-region is difficult.
- However, the reverse bias is favourable for minority electrons in p region and minority holes in n-region. Since the number of these carriers is very few, we see a **tiny reverse current**.



Definition
Rectifying action of pn junction is its property to allow **uni-directional** conduction.

- The diode equation can be shown to be

$$I = I_s \left[\exp\left(\frac{eV}{k_B T}\right) - 1 \right]$$

  where I_s is the reverse saturation current.

- The equation and the $I - V$ capture the non-linear nature of p-n junction.

- In the forward bias condition, it acts as a "closed switch" allowing large current to flow through it.

- In the reverse bias condition, it acts as an "open switch" allowing insignificantly

Problem
What causes majority carriers to flow at the moment when p-region and n-region are brought together? Why does this flow not continue until all carriers have recombined? [Hint for second part: Convince yourself that the built-in electric field opposes the diffusion "forces" for both the electrons and holes.]

Tough problems

Problem
Prove that the analysis of pn junction using potential energy barrier maps with the analysis using Fermi energy level. In other words, prove that $E_{Fn} - E_{Fp} = eV_{ext}$.

Test Your Understanding

1. What causes majority carriers to flow at the moment when p-region and n-region are brought together? Why does this flow not continue until all carriers have recombined? [Hint for second part: Convince yourself that the built-in electric field opposes the diffusion "forces" for both the electrons and holes.]
2. Explain the formation of **potential barrier** across the pn junction region.
3. Explain the formation of **depletion region** in the pn junction region. What is the typical value of depletion region width for commonly used pn junctions?
4. Draw band diagrams for symmetrically doped pn junction at
   - zero bias
   - forward bias
   - reverse bias
5. What is reverse saturation current in a diode?
6. Why is reverse saturation current independent of reverse bias?
7. What is reverse breakdown?

Technical terms

- metallurgical junction
- diffusion
- built-in electric field
- thermal equilibrium
- built-in potential barrier
- depletion region
- space charge region
- space charge width
- reverse saturation current
- reverse breakdown

M2U2 Plan

CFET

Quantum mechanics

QFET

Opto-electronics

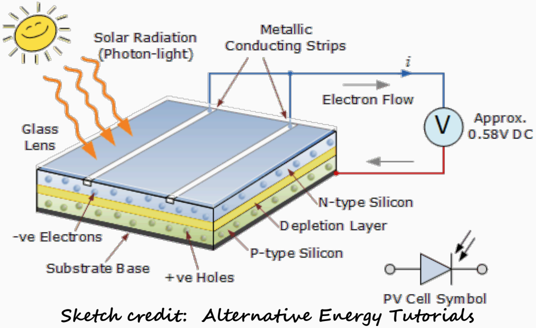
Optoelectronics

- Electronics is the study of controlling electrical signals.
- Any electrical signal is characterized by its current and voltage properties.
- Electronics is broadly classified into three areas –
    - **Microelectronics** – here we study switching electrical signals, which is useful to represent binary logic. This is necessary for computation. This is also called **digital electronics**. Since power is not a concern, signals have currents of the order of microamperes (hence the name microelectronics) and voltages of the order of few volts
    - **Power electronics** – here we study amplifying electrical signals to transfer energy. This is necessary for energy requirements of society as it is well known that electrical energy is efficient way of energy transfer. Signals have currents of the order of hundreds of amperes and voltages of the order of thousands of volts. This is also called **analog electronics**.
    - Opto-electronics [9] – here we harness the interaction between electron and photon. The advantages of light like its high velocity, free source of energy (Sun light), are combined with the advantages of electron like its easy manipulation to develop

Solar cell – construction

Definition

- Solar cell, also technically called the **photovoltaic cell** is an opto-electronic device based on pn junction that can generate electrical power when illuminated.
- A collection of cells in series and parallel combination is called a solar panel



Sketch of solar panel. Sketch taken from Alternative Energy Tutorials

Definition

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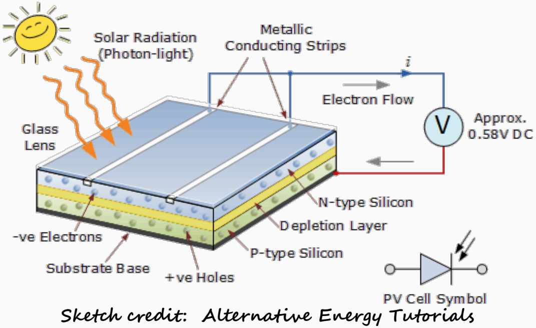


Sketch of solar panel. Sketch taken from Alternative Energy Tutorials

- The main aim of solar cell operation is to convert solar energy into electrical energy.
- Electrodes are deposited on both sides of junction. The top metal electrode is in the form of metal grid with fingers, which permits sunlight to pass through it.

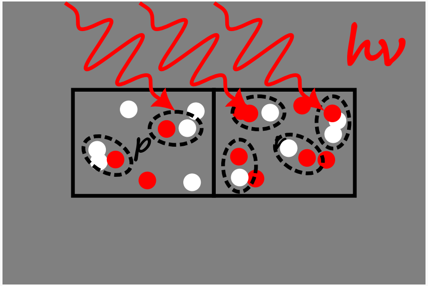
Definition
The working principle of solar cell involves three processes–

1. Generation of **excess** electron-hole pairs upon the illumination of light, and
2. Separation of electrons from holes **before** radiative recombination due to the built-in electric field.
3. Generation of voltage due to **accumulation** of excess separated charge carriers at the ends of the pn junction

- The solar cell pn junction is operated without bias i.e. zero bias.
- Upon illumination with photons of suitable energy, the photon are absorbed through the surface area. The covalent bonds of the lattice are broken releasing the free electrons and holes into the conduction band and valence band respectively.

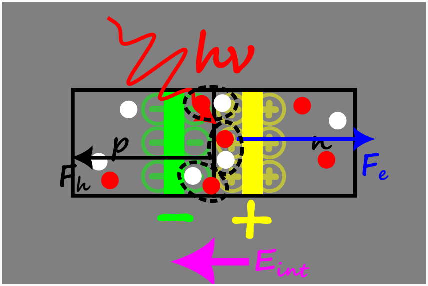
Problem
A material has bandgap of 5 eV. Can it be used as solar cell material? Why or why

Working principle – 1. Excess electron-hole pairs



- Upon illumination with photons with energy greater than the band gap, optical absorption takes place.
- This results in the generation of excess electron-hole pairs throughout the device.

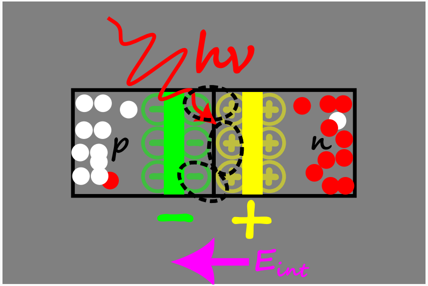
Key Insight
Optical absorption generates excess electron-hole pairs.



- At the depletion region also, the excess charge carriers are generated.
- However, before they undergo radiative recombination, the built-in electric field sweeps them out of the depletion.

Key Insight
At the depletion region, separation of excess charge carriers prevents radiative recombination.

Definition
Photovoltage, also called photo e.m.f. [10] is the voltage appearing across the device due to the separation of excess charge carriers.



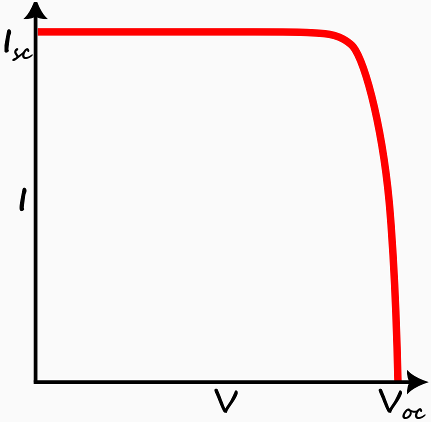
- The swept excess electrons and holes **accumulate** at the n region and p region edges respectively
- The accumulation of positive charge at the p-edge and negative charge at the n-edge leads to the generation of a voltage V_L.

- Since the source of voltage is illumination of photons, it is called **photovoltage**.

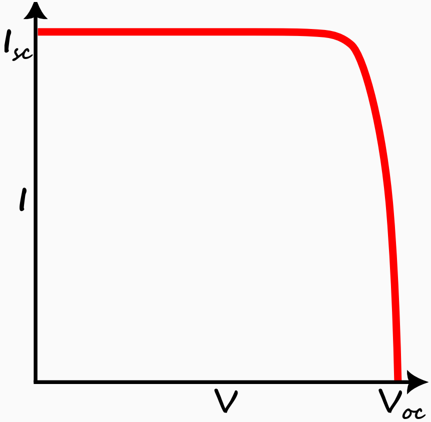
Key Insight
The photovoltage forward biases the pn junction.

[10] e.m.f. stands for electro motive force

Solar cell – $I - V$ characteristics



Solar cell – $I - V$ characteristics



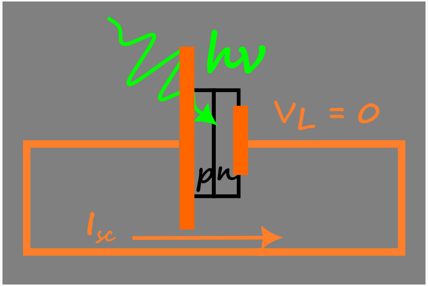
- From Kirchoff voltage law, the photo-voltage is given by
$$V_L = I_L R_L$$
- From diode $I - V$ characteristics, the forward current I_F is given by
$$I_F = I_s \left[\exp\left(\frac{eV_L}{k_B T}\right) - 1\right]$$

- Therefore the solar cell $I - V$ characteristics is given by
$$I_{\text{net}} = I_L - I_F$$
$$= I_L - I_s \left[\exp\left(\frac{eV_L}{\quad}\right) - 1\right]$$

Solar cell – short circuit current

Definition

Short circuit current is defined the current through solar cell when it is short circuited or zero load resistance is connected.



- When the load resistance is zero, the solar cell generates maximum current. This condition is called short circuit condition.
- At short circuit condition, $R_L = 0$, so the voltage drop across load is zero

$$V_L = I_L R_L = 0$$

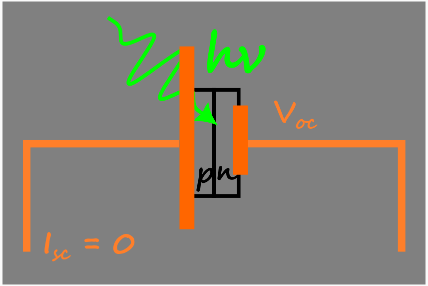
- The net current is the short circuit current I_{sc} and is given by

$$\begin{aligned} I_{sc} &= I_L - I_s\left(e^0 - 1\right) \\ &= I_L \end{aligned}$$

Solar cell – open circuit voltage

Definition

Open circuit voltage is defined the voltage across solar cell when it is open circuited or infinite load resistance is connected.



- When the load resistance is infinite, the solar cell generates maximum voltage. This condition is called open circuit condition.
- At open circuit condition, $R_L = \infty$, so the current is zero

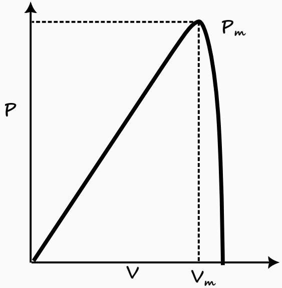
$$I_L = \frac{V_L}{R} = 0$$

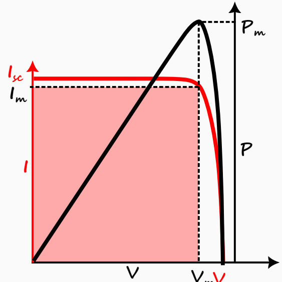
- The photovoltage is the open circuit voltage $V_{\rm oc}$ and is given by

$$I_L - I_s\left[\exp\left(\frac{eV_{\rm oc}}{k_B T}\right) - 1\right] = 0 \Rightarrow I_L = I_s\left[\exp\left(\frac{eV_{\rm oc}}{k_B T}\right) - 1\right]$$

$$\exp\left(\frac{eV_{\rm oc}}{k_B T}\right) = \frac{I_L}{I_s} + 1 \Rightarrow V_{\rm oc} = \left(\frac{k_B T}{e}\right)\ln\left(1 + \frac{I_L}{I_s}\right)$$

Solar cell – $P - V$ characteristics



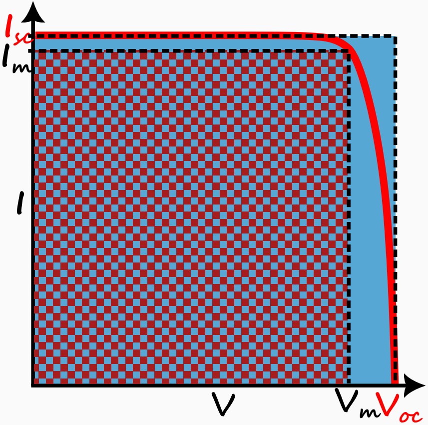


- The power generated in solar cell is completely consumed in the load resistance.
- If the photocurrent at V_m is I_m then the maximum power is

$$P_m = V_m I_m$$

- In $P - V$ curve, this is represented as as (V_m, P_m).
- The $I\&P - V$ curves show the mappings of $I - V$ and $P - V$ curves. In $P - V$ curve, maximum power is represented as as red rectangle.

Output electrical power – Fill factor



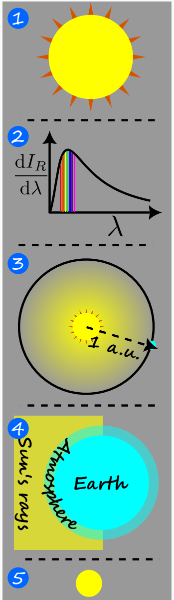
- From the $I - V$ characteristics of solar cell, it is evident that the maximum current generated cannot exceed the short circuit current I_{sc} and the maximum voltage generated cannot exceed the open circuit voltage V_{oc}.
- The theoretically possible maximum power that can be generated should be less than the product of I_{sc} and V_{oc}. In the $I - V$ curve, this is represented as blue rectangle.

- The ratio of experimentally possible maximum power to theoretically possible maximum power is called Fill Factor and is given by

$$FF = \frac{V_m I_m}{V_{oc} I_{sc}}$$

Key Insight

Input optical power – Solar irradiance



Definition
Solar irradiance is the solar power incident on the Earth's surface per unit area.

1. Sun as blackbody – Since sun is the major source of light energy, the input power is determined by the properties of sun.
2. Planck's blackbody radiation law – Sun is emitting photons of various wavelengths per unit time. Each photon with a given wavelength has an energy given by the Planck's law.
3. Sun -Earth separation – However, we are only receiving a fraction of the Sun's power.
4. Solar irradiance outside atmosphere – It is $1360\,\mathrm{W\,m^{-2}}$.
5. Solar irradiance at Earth's surface – It is nearly $1000\,\mathrm{W\,m^{-2}}$ at noon due to losses in atmosphere.

Practical considerations – Solar cell efficiency

Definition
Conversion efficiency is defined as ratio of output electrical power to input optical power.

- To harness solar energy, the following practical limitations need to be considered.
- All the power cannot be harnessed as the optical absorption by a solar cell has a cut off wavelength due to the energy band gap.
- The photons with energy greater than energy band gap also generate electron-hole pairs. However, the difference of photon energy and band gap is dissipated as heat.
- The sum total of all these considerations is that a fraction of incident optical power is converted into electrical power and is given by the conversion efficiency η as

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100$$

Chandrayaan 2 misson at glance [Click for video]

Image credits: ISRO

Problem

Test Your Understanding

1. Explain the working of solar cell.
2. Draw and explain the $I - V$ characteristics of a solar cell.
3. A material has bandgap of 5 eV. Can it be used as solar cell material? Why or why not?
4. An alien ship crashlanded onto Earth's surface. From the debris, the solar panel used for power generation is found to have a band gap of 5 eV. Estimate the temperature of the alien Sun.

Technical terms

- optical absorption
- generation of excess charge carriers
- conversion efficiency
- fill factor
- open-circuit voltage
- short-circuit current

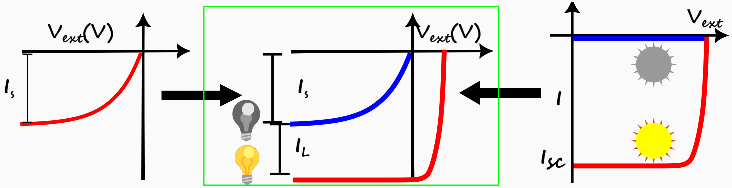
M2U2 Plan

Photo-diode

Definition
A photo-diode is an opto-electronic device that is used to sense the presence of light.

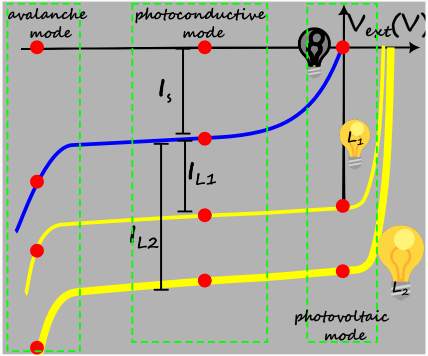
- A photo-diode is a pn junction diode that operates under **reverse bias**.
- Due to the reverse bias, the depletion region width increase. This leads to decrease in junction capacitance.

Photo-diode – $I - V$ characteristics



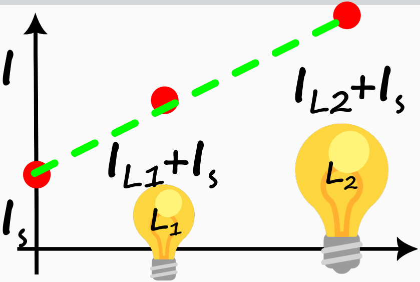
- The $I - V$ characteristics of photodiode can be considered as $I - V$ characteristics of solar cell under forward bias and $I - V$ characteristics of pn junction under reverse bias.
- The reverse bias generates reverse current that saturates to reverse saturation current I_s at high bias.
- In addition to the reverse current, if the photons are incident on the diode there is an additional photocurrent I_L.
- Thus the external bias is negative and the net current is negative and the $I - V$ characteristics is given by

Photo-diode – Modes of operation



- The photo diode has three modes of operation
  1. photoconductive mode – reverse bias
  2. zero bias mode – reverse bias voltage is set to zero
  3. avalanche mode – **very** high reverse bias
- In every mode of operation, there are two states of operation

Photoconductive mode



- Photo diode is under reverse bias.
- At zero light intensity, there is a non zero current due to the reverse saturation current. This is called **dark current**.
- Response of photodiode is linear. In this mode, photodiode is a linear sensor.

- Due to reverse bias, the depletion region width increases. Since the space charge region acts as a parallel plate capacitor, the junction capacitance C_{junction} is given by

$$C_{\text{junction}} = \frac{\epsilon A}{W}$$

where A is the cross sectional area of the depletion region and W is the depletion region width.
- Due to reduction in junction capacitance, photodiode is used for high frequency

M2U2 Plan

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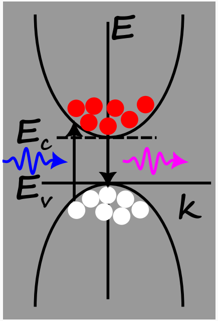
Opto-electronics

Direct band gap semiconductor

Definition
A direct band gap semiconductor is defined as a semiconductor with the top of valence band and bottom of conduction band at the same value of wave vector.



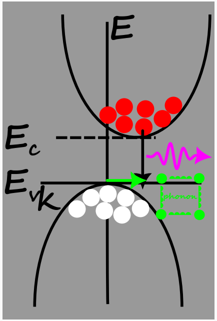
- Consider electron moves from E_c to E_v.
- In the radiative recombination process,
  - Energy of electron is transferred to energy of photon.
  - Initial momentum of electron is equal to final momentum.
  - Therefore, both the energy and momentum of the (electron + hole + photon) system is **conserved**.
  - **three** particles are involved

$$\mathrm{e}^- + \mathrm{h}^+ \rightarrow \text{photon}$$

Key Insight
Radiative type of recombination is the dominant mechanism in a direct bandgap semiconductor.

Indirect band gap semiconductor

Definition
An indirect band gap semiconductor is defined as a semiconductor with the top of valence band and bottom of conduction band at the **different** values of wave vector.



- Consider electron moves from E_c to E_v.
- In the radiative recombination process,
  - Energy of electron is transferred to energy of photon.
  - Initial momentum of electron is not equal to final momentum.
  - Therefore, energy of the (electron + hole + photon) system is **conserved** but the momentum is not conserved.

- To conserve momentum, a **phonon assisted radiative recombination** occurs where difference in momentum is supplied by phonon and **four** particles are involved

Direct vs indirect bandgap semiconductor

SNo.	Property	Direct	Indirect
1	k at E_C vs k at E_v	equal	unequal
2	Recombination mechanism	Radiative	Non-radiative
3	Particle emitted	Photon	Phonon
4	Periodic table	Compound s.c. like GaAs, InP	Elemental s.c. like Si, Ge
5	Applications	LED, laser diode	transistors, diodes

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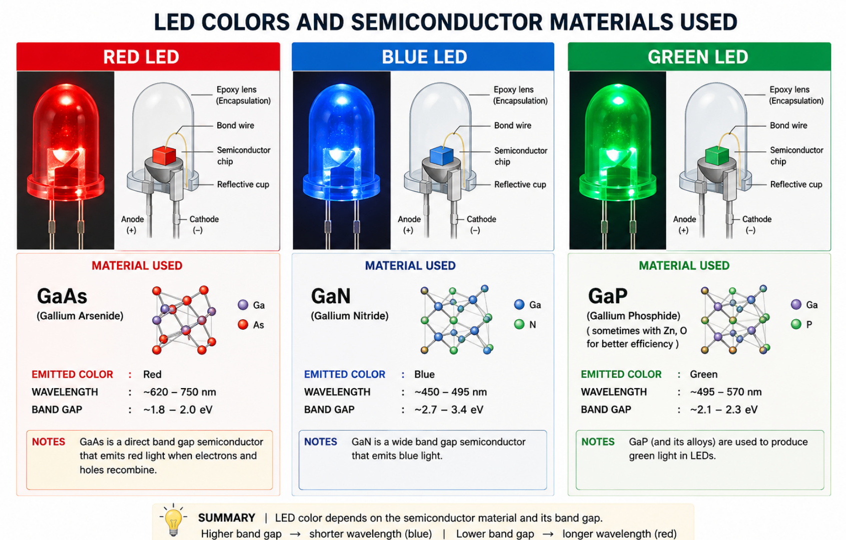
Opto-electronics

Light Emitting Diode

Definition
Light Emitting Diode (LED) is the opto-electronic device based on radiative recombination of charge carriers.

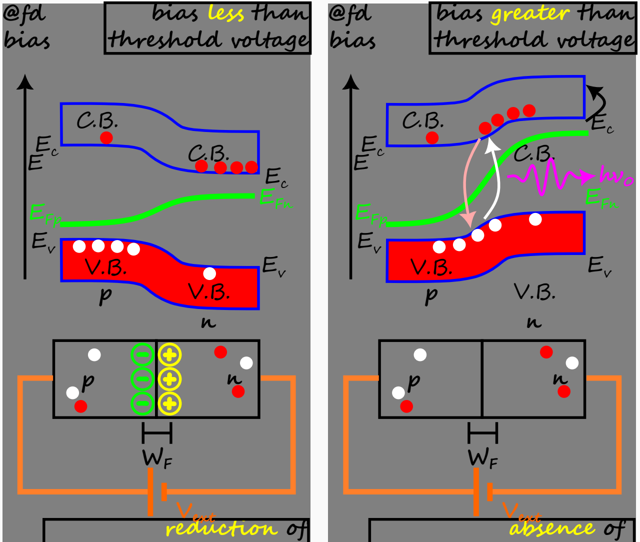
- LED is based on pn junction diode under forward bias.
- Under forward bias, the electrons and holes crossing the depletion region undergo radiative recombination leading to generation of light.
- From the perspective of electroluminescence, it is useful to use a semiconductor with a direct band gap semiconductor.

LED – construction

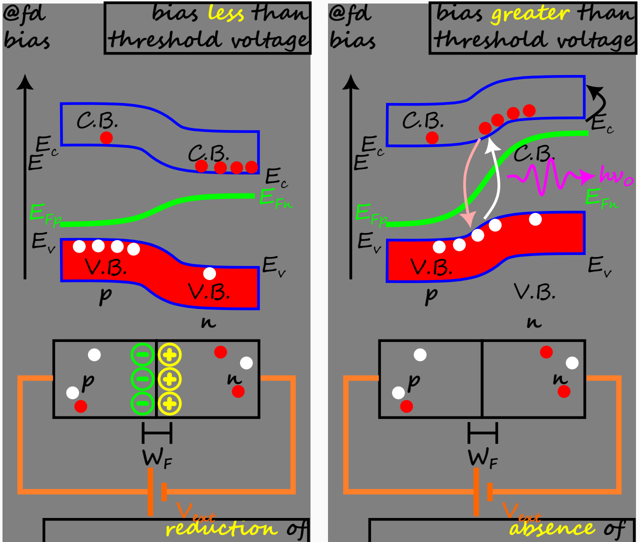


- New materials AlGaInP for Red, $In_xGa_{1-x}N$ for Green, and $In_xGa_{1-x}N$ for Blue.

LED – working principle



LED – working principle



- LED works on the principle of **electroluminescence**.
- Below threshold voltage, potential energy barrier is reduced, and depletion region width is reduced.
- Above threshold voltage, potential energy barrier is removed, and depletion region width is removed. The electrons and holes move towards the junction, and **recombine radiatively**

LED – optical emission

- From the Planck's quantum hypothesis, the wavelength can be related to the energy of photon by

$$\begin{aligned} E &= h\nu, \\ &= \frac{hc}{\lambda}, \\ E\,(\text{in eV}) &= \frac{1.24}{\lambda\,(\text{in µm})} \end{aligned} \tag{A}$$

Problem
Prove Equation. A.

Problem
Find the energy of red photon with wavelength 700 nm.

Problem
Find the energy of blue photon with wavelength 400 nm.

LED – applications

- Night lighting
- Traffic signals
- Optical fibre communications
- Medical imaging

Summary of opto-electronic devices

- From the perspective of bias

Reverse bias	Zero bias	Forward bias
Photo-diode	Solar cell	LED

- From the perspective of depletion region width

Larger width	Nominal width	Smaller width
Low current	Nominal current	Large current

- From the perspective of power consumption

Low Power	Nominal Power	High Power
Consumption	Generation	Consumption

- From the perspective of capacitance

Low Capacitance	Nominal Capacitance	High Capacitance
High frequency	–	–

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Opto-electronics

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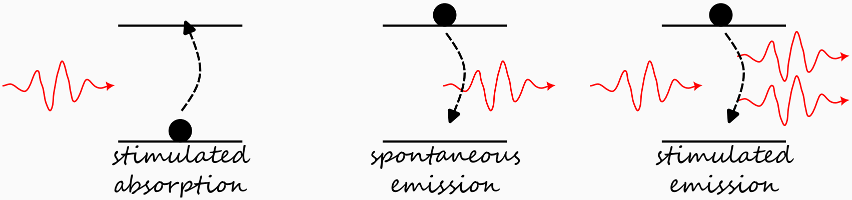
Opto-electronics

- L.A.S.E.R is acronym for Light Amplification by Stimulated Emission of Radiation.
- Einstein predicted the possibility of lasing in 1916 by introducing concept of stimulated emission.



- Light interacts with matter by three quantum processes
  - Stimulated absorption
  - Spontaneous emission
  - Stimulated emission

Laser – Three Quantum Processes (contd...)

Definition
Stimulated absorption occurs when electron transitions from lower energy level to higher energy level upon the absorption of photon.

$$\mathrm{e}^- + \mathrm{h}\nu \longrightarrow \mathrm{e}^*$$

Definition
Spontaneous emission occurs when electron transitions from higher energy level to lower energy level by the emission of photon.

$$\mathrm{e}^* \longrightarrow \mathrm{e}^- + \mathrm{h}\nu$$

Definition
Stimulated emission occurs when electron transitions from higher energy level to lower energy level in the presence of a photon by the emission of additional photon.

Laser – Conditions and requirements for lasing action

From the Einstein's theory [11] of light-matter interaction, the conditions for lasing action are –

- high photon density,
- large lifetime of excited levels, and
- high population of excited levels to achieve lasing action.

To satisfy these three conditions for lasing action, the requirement of lasing action are –

1. Population inversion
2. Meta-stable energy level
3. Radiation confinement

[11] Refer to Chapter 44 Avadhanulu for a discussion on this theory.

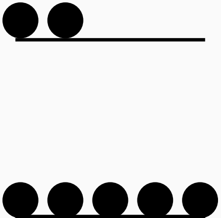
Thermal equilibrium and normal state

Definition
Maxwell-Boltzmann statistics says the probability of occupying higher energy level is lower than the probability of occupying lower energy level. The probability of occupying energy level E is proportional to

$$P_{MB} \propto \exp\left(-\frac{E}{k_B T}\right)$$

Definition
Normal state is defined as the state under thermal equilibrium.

- Under the conditions of thermal equilibrium, the occupancy of energy levels is governed by Maxwell Boltzmann statistics.



Problem
Find the ratio of populations of two states in a He- Ne laser

Laser – components

To satisfy the above three requirements of lasing action, we need three components to any laser –

1. Active medium
2. Optical pump
3. Optical resonant cavity

Laser – characteristics

Laser light 	Ordinary light 
Unidirectional	Non-directional
Coherent	Incoherent
Monochromatic	Non-monochromatic
High intensity	Low intensity
Low pulse width	High pulse width

- Coherent means photons in the beam have same phase
- Monochromatic beam consists of photons of single color/wavelength.

Diode laser – Construction

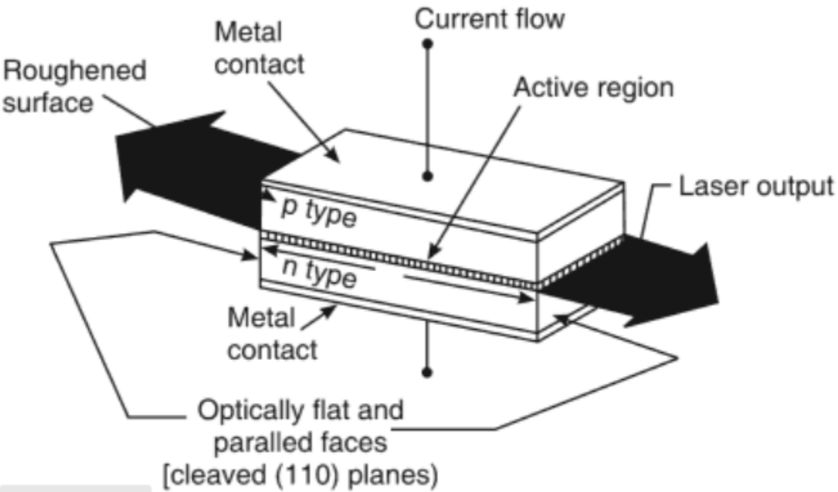
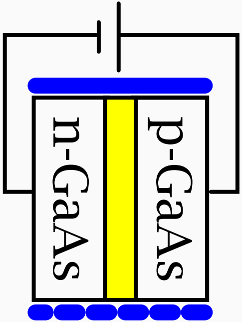
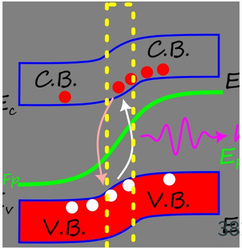


Fig. 44.24:

Schematic of homojunction diode laser

- In a diode laser
  - Depletion region is the active medium,
  - Forward biasing is the pump, and
  - Polished faces of the GaAs crystal are the optical resonator.
- Since the forward biasing populates the electrons in the n region and holes in the p region, when they meet at the depletion region, the active medium is already in the lasing state.
- Therefore, there is no need for a meta-stable state.





Diode laser – applications

- Industrial precision cutting
- Lasik surgery
- LiDAR (Light Imaging, Detection and Ranging) technology for distance measurement
- Optical fibre communication technology

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CFET

Quantum mechanics

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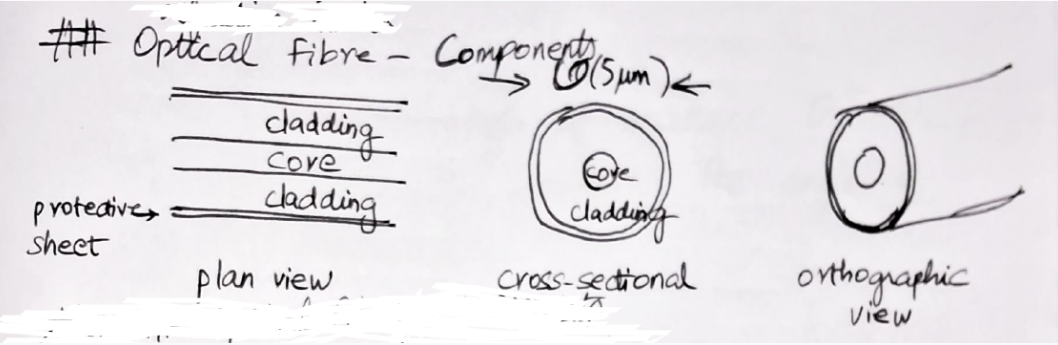
Opto-electronics

Optical fiber – light pipe

- Optical fiber is an opto-electronic device that is the backbone of modern communication systems.
- Useful for long distance communication.
- Harnesses the speed of light and property of light propagation across media with different refractive indices.


Optical fibre – Components
(5μm)
cladding
core
cladding
protective sheet
plan view
core
cladding
cross-sectional
orthographic view

Optical fiber – working principle

- The speed of light varies in different media.
- It is highest in vacuum and is given by $v = c$.
- In other medium, it is quantified by the refractive index n, so that the velocity of light in the medium is

$$v = \frac{c}{n}.$$

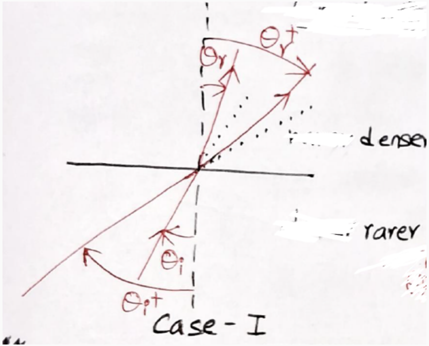
- At the interface of media, speed of light changes leading to two cases.
  - Case I: Light propagates from rarer to denser medium.
  - Case II: Light propagates from denser to rarer medium.

medium	n
air	1
water	1.33
glass	1.5

Table 9: Refractive indices of commonly available media.

Case I: Rarer to denser medium

Consider light is incident at an angle of incidence θ_i with respect to normal.

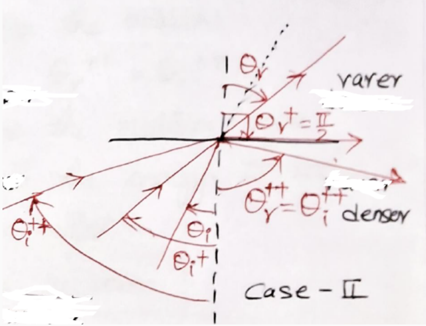


- Since light is faster in rarer medium, and slower in denser medium, it bends towards the normal. Therefore, it refracts.
- Angle of refraction is less than angle of incidence,

$$\theta_r < \theta_i.$$

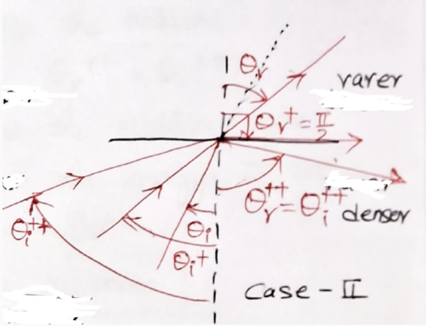
- Similarly, for larger angle of incidence θ_i^+, light refracts with angle of refraction $\theta_r^+ < \theta_i^+$.

Case II: Denser to rarer medium



- Since light is slower in denser medium and faster in rarer medium, it bends away from the normal.
- Case (i)Angle of refraction is more than angle of incidence,
$$\theta_r > \theta_i.$$
- At a critical angle of incidence $\theta_i^+ = \theta_c$, the light grazes the interface. The angle of refraction is
$$\theta_r^+ = \frac{\pi}{2}$$

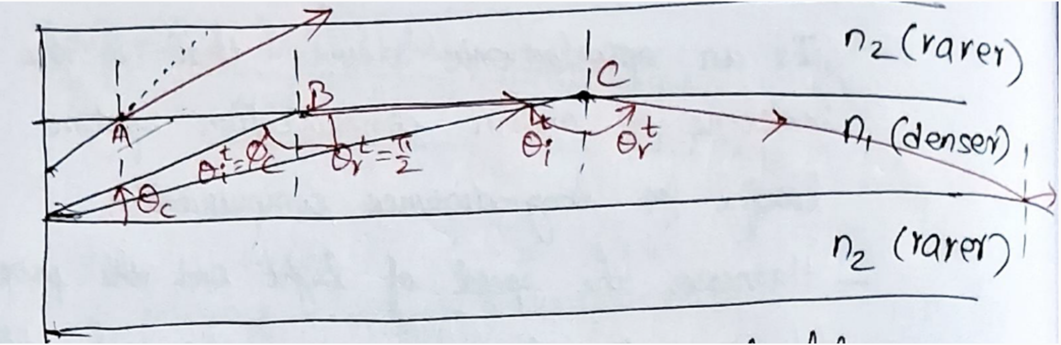
Total internal reflection



- Case (iii) For angle of incidence $\theta_i^{++} > \theta_c$, the light reflects back into the medium. The angle of reflection

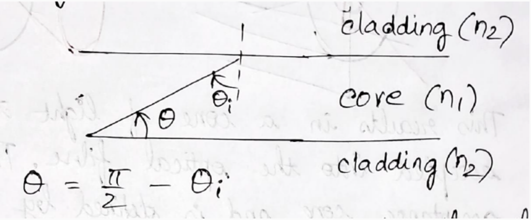
$$\theta_r^{++} = \theta_i^{++}.$$

- Since the light enters the medium again with reflection and all the light energy is within the medium, this condition is called **Total Internal Reflection** (TIR).
- TIR is required for light containment within the optical fiber. This possible if refractive index of core is greater than refractive index of cladding.



- Light is at one end of the optical fiber.
- Not all light can be contained within the light pipe (optical fiber).

Definition
Angle of propagation is defined as the angle between incident ray and core-cladding interface.



Acceptance angle

- Before the light enters the core medium, it has to encounter the interface between the core and the ambient [12] medium.
- If the refractive index of ambient medium is n_3, then for entering the core medium, the refractive index of ambient must be lesser than refractive index of core.
- For convenience, let us consider $n_3 = 1$. Therefore, the angle of incidence at the ambient-core interface needs to be less than the so called acceptance angle so that the refracted light is within the critical angle of propagation.
- At the acceptance angle of incidence θ_0,

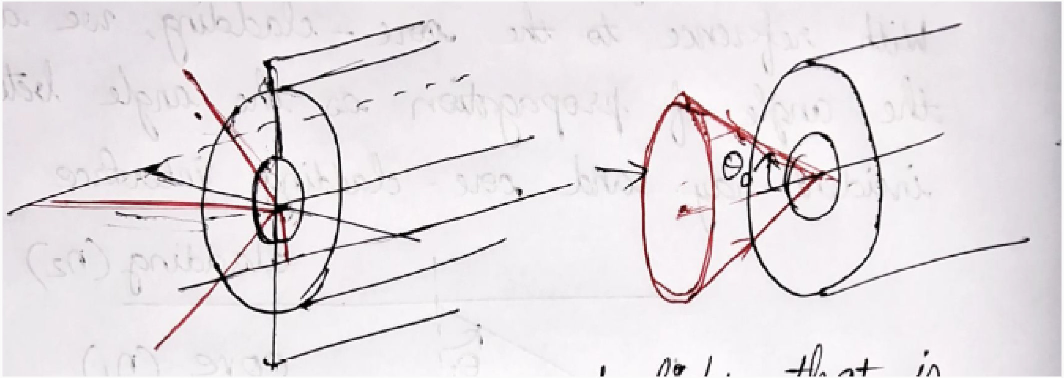
$$\frac{\sin\theta_0}{\sin\theta_0} = \frac{n_1}{n_3} = n_1$$

$$\theta_0 = \sin^{-1}\sqrt{n_1^2 - n_2^2}$$

[12] ambient means outside or commonly occurring, in our case it is usually air.

Acceptance cone

- We have limited our previous analysis to the "top" and "bottom" interface.
- Due to the cylindrical nature of optical fiber, the analysis is applicable to all planes passing through the axis.



This results in a cone of light that is accepted into the optical fiber. This is called

Optical fiber based communication

- The industry has defined new quantities to quantify an optical fibre –
    1. numerical aperture
    2. fractional refractive index

Fractional refractive index

Definition
Fractional refractive index is defined as relative difference in the refractive indices of core and cladding.

- The fractional refractive index is given by

$$\Delta = \frac{n_1 - n_2}{n_1}.$$

- In order to guide light effectively through a fiber, *Delta* $\ll 1$. Typically, *Delta* is of the order of 0.01.

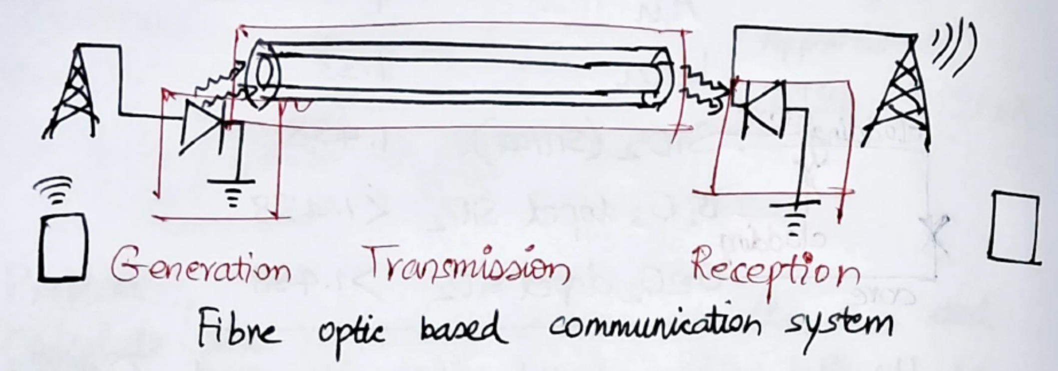
Numerical aperture

Definition
Numerical aperture is defined as the sine of the acceptance angle

$$
\begin{aligned}
\text{N.A.} &= \sin\theta_0, \\
&= \sin\left(\sin^{-1}\sqrt{n_1^2 - n_2^2}\right). \\
&= \sqrt{n_1^2 - n_2^2}, \\
&= \sqrt{(n_1 + n_2)(n_1 - n_2)}, \\
&= \sqrt{\left(\frac{n_1 + n_2}{2}\right)\left(\frac{n_1 - n_2}{n_1}\right) \cdot 2n_1}, \\
&\simeq \sqrt{n_1 \Delta 2 n_1}, \\
&= n_1\sqrt{2\Delta}.
\end{aligned}
$$

Optical fiber based communication

- Optical fiber is the backbone of modern communication system.



End of M2U3